Tutorial 12: nMDS, PERMANOVA and SIMPER

ENVX2001 – Applied Statistical Methods

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Semester 1

Aim of this tutorial

In the previous tutorial we used **hierarchical clustering** to ask whether sites with similar bird communities group together.

In this tutorial we use the **same Mac Nally bird data**, but analyse it using three related multivariate methods:

1. **nMDS** — to visualise similarity among bird communities.
2. **ANOSIM** — to test whether communities differ among habitat groups.
3. **PERMANOVA** — to test whether habitat type explains multivariate differences in bird assemblages.
4. **SIMPER** — to identify which bird species contribute most to differences among habitat groups.

The data are based on Mac Nally's study of forest and woodland birds along a 250 km transect in south-eastern Australia. The paper examined bird assemblages across contrasting forest and woodland habitats, including river red gum/grey box woodlands, foothill forests, montane forests, red ironbark forests, manna gum woodlands and coastal banksia woodland.

Mac Nally used multivariate approaches to understand habitat structure, bird occurrence and abundance across sites. Here, we focus on the **bird community matrix**: rows are sites and columns are bird species.

Load packages

CODE

```r
library(vegan)
```

OUTPUT

```
Loading required package: permute
```

CODE

```r
library(MASS)
library(ggplot2)
library(dplyr)
```

OUTPUT

```

Attaching package: 'dplyr'
```

OUTPUT

```
The following object is masked from 'package:MASS':

    select
```

OUTPUT

```
The following objects are masked from 'package:stats':

    filter, lag
```

OUTPUT

```
The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

Load the data

The dataset contains:

- `SITE` = site name
- `HABITAT` = habitat category
- bird species columns = abundance/density estimates for each bird species

CODE

```r
Birds <- read.csv("data/macnally.csv", header = TRUE)


head(Birds)
```

OUTPUT

```
         SITE      HABITAT V1GST V2EYR V3GF V4BTH V5GWH V6WTTR V7WEHE V8WNHE
1  Reedy Lake        Mixed   3.4   0.0  0.0   0.0   0.0    0.0    0.0   11.9
2  Pearcedale Gippsland Ma   3.4   9.2  0.0   0.0   0.0    0.0    0.0   11.5
3     Warneet Gippsland Ma   8.4   3.8  0.7   2.8   0.0    0.0   10.7   12.3
4  Cranbourne Gippsland Ma   3.0   5.0  0.0   5.0   2.0    0.0    3.0   10.0
5 Lysterfield        Mixed   5.6   5.6 12.9  12.2   9.5    2.1    7.9   28.6
6    Red Hill        Mixed   8.1   4.1 10.9  24.5   5.6    6.7    9.4    6.7
  V9SFW V10WBSW V11CR V12LK V13RWB V14AUR V15STTH V16LR V17WPHE V18YTH V19ER
1   0.4     0.0   1.1   3.8    9.7    0.0     0.0   4.8    27.3      0   5.1
2   8.3    12.6   0.0   0.5   11.6    0.0     0.0   3.7    27.6      0   2.7
3   4.9    10.7   0.0   1.9   16.6    2.3     2.8   5.5    27.5      0   5.3
4   6.9    12.0   0.0   2.0   11.0    1.5     0.0  11.0    20.0      0   2.1
5   9.2     5.0  19.1   3.6    5.7    8.8     7.0   1.6     0.0      0   1.4
6   0.0     8.9  12.1   6.7    2.7    0.0    16.8   3.4     0.0      0   2.2
  V20PCU V21ESP V22SCR V23RBFT V24BFCS V25WAG V26WWCH V27NHHE V28VS V29CST
1      0    0.0    0.0     0.0     0.6    1.9       0     0.0   0.0    1.7
2      0    3.7    0.0     1.1     1.1    3.4       0     6.9   0.0    0.9
3      0    0.0    0.0     0.0     1.5    2.1       0     3.0   0.0    1.5
4      0    2.0    0.0     5.0     1.4    3.4       0    32.0   0.0    1.4
5      0    3.5    0.7     0.0     2.7    0.0       0     6.4   0.0    0.0
6      0    3.4    0.0     0.7     2.0    0.0       0     2.2   5.4    0.0
  V30BTR V31AMAG V32SCC V33RWH V34WSW V35STP V36YFHE V37WHIP V38GAL V39FHE
1   12.5     8.6   12.5    0.6    0.0    4.8     0.0       0    4.8   26.2
2    0.0     0.0    0.0    2.3    5.7    0.0     1.1       0    0.0    0.0
3    0.0     0.0    0.0    1.4   24.3    3.1    11.7       0    0.0    0.0
4    0.0     0.0    0.0    0.0   10.0    4.0     0.0       0    2.8    0.0
5    0.0     0.0    0.0    7.0    0.0    0.0     6.1       0    0.0    0.0
6    0.0     0.0    0.0    6.8    0.0    0.0     0.0       0    0.0    0.0
  V40BRTH V41SPP V42SIL V43GCU V44MUSK V45MGLK V46BHHE V47RFC V48YTBC V49LYRE
1       0    0.0    0.0    0.0    13.1     1.7     1.1      0       0       0
2       0    1.1    0.0    0.0     0.0     0.0     0.0      0       0       0
3       0    4.6    0.0    0.0     0.0     0.0     0.0      0       0       0
4       0    0.8    0.0    0.0     0.0     1.4     0.0      0       0       0
5       0    5.4    0.0    0.0     0.0     0.0     0.0      0       0       0
6       0    3.4    2.7    1.4     0.0     0.0     0.0      0       0       0
  V50CHE V51OWH V52TRM V53MB V54STHR V55LHE V56FTC V57PINK V58OBO V59YR V60LFB
1      0      0     15     0       0      0    0.0       0    0.0     0    2.9
2      0      0      0     0       0      0    2.3       0    0.0     0    0.0
3      0      0      0     0       0      0    0.0       0    0.0     0    0.0
4      0      0      0     1       0      0    0.0       0    0.0     0    0.0
5      0      0      0     0       0      0    2.1       0    1.4     0    0.0
6      0      0      0     0       0      0    0.0       0    0.0     0    0.0
  V61SPW V62RBTR V63DWS V64BELL V65LWB V66CBW V67GGC V68PIL V69SKF V70RSL
1      0       0    0.4     0.0      0      0      0      0    1.9    6.7
2      0       0    0.0     0.0      0      0      0      0    0.0    0.0
3      0       0    3.5     0.0      0      0      0      0    0.0    0.0
4      0       0    5.5     0.0      4      0      0      0    0.0    0.8
5      0       0    0.0    22.1      0      0      0      0    0.0    0.0
6      0       0    0.0     0.0      0      0      0      0    0.0    0.0
  V71PDOV V72CRP V73JW V74BCHE V75RCR V76GBB V77RRP V78LLOR V79YTHE V80RF
1       0      0     0       0      0    0.0    4.8       0       0   0.0
2       0      0     0       0      0    0.0    0.0       0       0   0.0
3       0      0     0       0      0    0.0    0.0       0       0   0.0
4       0      0     0       0      0    0.0    0.0       0       0   0.0
5       0      0     0       0      0    0.7    0.0       0       0   0.0
6       0      0     0       0      0    0.0    0.0       0       0   1.4
  V81SHBC V82AZKF V83SFC V84YRTH V85ROSE V86BCOO V87LFC V88WG V89PCOO V90WTG
1     0.0       0    0.0       0       0       0    0.0     0     1.9      0
2     0.0       0    0.0       0       0       0    0.0     0     0.0      0
3     1.4       0    0.0       0       0       0    1.8     0     0.0      0
4     0.0       0    0.0       0       0       0    0.0     0     0.0      0
5     0.7       0    0.0       0       0       0    0.0     0     0.0      0
6     0.0       0    3.4       0       0       0    0.0     0     0.0      0
  V91NMIN V92NFB V93DB V94RBEE V95HBC V96DF V97PCL V98FLAME V99WWT V100WBWS
1     0.2      0     0       0      0     0    9.1        0      0        0
```

```
2     0.0      0     0       0      0     0    0.0        0      0        0
3     5.8      0     0       0      0     0    0.0        0      0        0
4     3.1      0     0       0      0     0    0.0        0      0        0
5     0.0      0     0       0      0     0    0.0        0      0        0
6     0.0      0     0       0      0     0    0.0        0      0        0
  V101LCOR V102KING
1        0        0
2        0        0
3        0        0
4        0        0
5        0        0
6        0        0
```

CODE

```
str(Birds)
```

OUTPUT

```
'data.frame':	37 obs. of  104 variables:
 $ SITE    : chr  "Reedy Lake" "Pearcedale" "Warneet" "Cranbourne" ...
 $ HABITAT : chr  "Mixed" "Gippsland Ma" "Gippsland Ma" "Gippsland Ma" ...
 $ V1GST   : num  3.4 3.4 8.4 3 5.6 8.1 8.3 4.6 3.2 4.6 ...
 $ V2EYR   : num  0 9.2 3.8 5 5.6 4.1 7.1 5.3 5.2 1.2 ...
 $ V3GF    : num  0 0 0.7 0 12.9 10.9 6.9 11.1 8.3 4.6 ...
 $ V4BTH   : num  0 0 2.8 5 12.2 24.5 29.1 28.2 18.2 6.5 ...
 $ V5GWH   : num  0 0 0 2 9.5 5.6 4.2 3.9 3.8 2.3 ...
 $ V6WTTR  : num  0 0 0 0 2.1 6.7 2 6.5 4.2 5.2 ...
 $ V7WEHE  : num  0 0 10.7 3 7.9 9.4 7.1 2.6 2.8 0.6 ...
 $ V8WNHE  : num  11.9 11.5 12.3 10 28.6 6.7 27.4 10.9 9 3.6 ...
 $ V9SFW   : num  0.4 8.3 4.9 6.9 9.2 0 13.1 3.1 3.8 3.8 ...
 $ V10WBSW : num  0 12.6 10.7 12 5 8.9 2.8 8.6 5.6 3 ...
 $ V11CR   : num  1.1 0 0 0 19.1 12.1 0 9.3 14.1 7.5 ...
 $ V12LK   : num  3.8 0.5 1.9 2 3.6 6.7 2.8 3.8 3.2 2.4 ...
 $ V13RWB  : num  9.7 11.6 16.6 11 5.7 2.7 2.4 0.6 0 0.6 ...
 $ V14AUR  : num  0 0 2.3 1.5 8.8 0 2.8 1.3 0 0 ...
 $ V15STTH : num  0 0 2.8 0 7 16.8 13.9 10.2 12.2 11.3 ...
 $ V16LR   : num  4.8 3.7 5.5 11 1.6 3.4 0 0 0.6 5.8 ...
 $ V17WPHE : num  27.3 27.6 27.5 20 0 0 16.7 0 0 0 ...
 $ V18YTH  : num  0 0 0 0 0 0 0 0 0 9.6 ...
 $ V19ER   : num  5.1 2.7 5.3 2.1 1.4 2.2 0 1.2 1.3 2.3 ...
 $ V20PCU  : num  0 0 0 0 0 0 0 0 2.8 2.9 ...
 $ V21ESP  : num  0 3.7 0 2 3.5 3.4 5.5 5.1 7.1 0.6 ...
 $ V22SCR  : num  0 0 0 0 0.7 0 0 0 0 3 ...
 $ V23RBFT : num  0 1.1 0 5 0 0.7 0 0.7 1.9 0 ...
 $ V24BFCS : num  0.6 1.1 1.5 1.4 2.7 2 3.6 0 0.6 1.2 ...
 $ V25WAG  : num  1.9 3.4 2.1 3.4 0 0 0 0 0 0 ...
 $ V26WWCH : num  0 0 0 0 0 0 0 0 0 9.8 ...
 $ V27NHHE : num  0 6.9 3 32 6.4 2.2 5.6 0 0 0 ...
 $ V28VS   : num  0 0 0 0 0 5.4 5.6 1.9 4.2 5.1 ...
 $ V29CST  : num  1.7 0.9 1.5 1.4 0 0 4.6 0 0 0 ...
 $ V30BTR  : num  12.5 0 0 0 0 0 0 0 0 0 ...
 $ V31AMAG : num  8.6 0 0 0 0 0 0 0 0 0 ...
 $ V32SCC  : num  12.5 0 0 0 0 0 0 0 0 0 ...
 $ V33RWH  : num  0.6 2.3 1.4 0 7 6.8 7.3 9.1 4.5 6.3 ...
 $ V34WSW  : num  0 5.7 24.3 10 0 0 3.6 0 0 0 ...
 $ V35STP  : num  4.8 0 3.1 4 0 0 2.4 0 0 3.5 ...
 $ V36YFHE : num  0 1.1 11.7 0 6.1 0 0 0 0 2.3 ...
 $ V37WHIP : num  0 0 0 0 0 0 0 2 3.2 0 ...
 $ V38GAL  : num  4.8 0 0 2.8 0 0 0 0 0 0 ...
 $ V39FHE  : num  26.2 0 0 0 0 0 0 0 0 0 ...
 $ V40BRTH : num  0 0 0 0 0 0 0 0 0 6 ...
 $ V41SPP  : num  0 1.1 4.6 0.8 5.4 3.4 0 2 2.6 4.2 ...
 $ V42SIL  : num  0 0 0 0 0 2.7 1.2 0 4.9 1.2 ...
```

```
 $ V43GCU  : num  0 0 0 0 0 1.4 0 2.6 1.3 0 ...
 $ V44MUSK : num  13.1 0 0 0 0 0 0 0 0 0 ...
 $ V45MGLK : num  1.7 0 0 1.4 0 0 0 0 0 0 ...
 $ V46BHHE : num  1.1 0 0 0 0 0 0 0 0 0 ...
 $ V47RFC  : num  0 0 0 0 0 0 0 0 0 0 ...
 $ V48YTBC : num  0 0 0 0 0 0 0 1.9 2.6 0 ...
 $ V49LYRE : num  0 0 0 0 0 0 0 0 0.6 0 ...
 $ V50CHE  : num  0 0 0 0 0 0 0 0 0.6 0 ...
 $ V51OWH  : num  0 0 0 0 0 0 0 0 0 0 ...
 $ V52TRM  : num  15 0 0 0 0 0 0 0 0 0 ...
 $ V53MB   : num  0 0 0 1 0 0 3.6 0 0 1.2 ...
 $ V54STHR : num  0 0 0 0 0 0 0 0 0 0 ...
 $ V55LHE  : num  0 0 0 0 0 0 0 0 0 0 ...
 $ V56FTC  : num  0 2.3 0 0 2.1 0 0 2.6 2.6 1.7 ...
 $ V57PINK : num  0 0 0 0 0 0 0 0 0 0 ...
 $ V58OBO  : num  0 0 0 0 1.4 0 0 0 0 1.2 ...
 $ V59YR   : num  0 0 0 0 0 0 0 0 0 0 ...
 $ V60LFB  : num  2.9 0 0 0 0 0 0 0 0 0 ...
 $ V61SPW  : num  0 0 0 0 0 0 0 0 0 0 ...
 $ V62RBTR : num  0 0 0 0 0 0 0 0 0 0.6 ...
 $ V63DWS  : num  0.4 0 3.5 5.5 0 0 1.8 0 0 0 ...
 $ V64BELL : num  0 0 0 0 22.1 0 0 0 0 0 ...
 $ V65LWB  : num  0 0 0 4 0 0 0 0 0 0 ...
 $ V66CBW  : num  0 0 0 0 0 0 0 0 0 0.6 ...
 $ V67GGC  : num  0 0 0 0 0 0 0 0 1.3 0 ...
 $ V68PIL  : num  0 0 0 0 0 0 0 0 1.3 0 ...
 $ V69SKF  : num  1.9 0 0 0 0 0 0 0 0 2.3 ...
 $ V70RSL  : num  6.7 0 0 0.8 0 0 0 0 0 0 ...
 $ V71PDOV : num  0 0 0 0 0 0 0 0 0 0 ...
 $ V72CRP  : num  0 0 0 0 0 0 0 0 0 0 ...
 $ V73JW   : num  0 0 0 0 0 0 0 0 0 0 ...
 $ V74BCHE : num  0 0 0 0 0 0 0 0 0 0 ...
 $ V75RCR  : num  0 0 0 0 0 0 0 0 0 0 ...
 $ V76GBB  : num  0 0 0 0 0.7 0 0 0 0.6 0 ...
 $ V77RRP  : num  4.8 0 0 0 0 0 0 0 0 0 ...
 $ V78LLOR : num  0 0 0 0 0 0 0 0 0 0 ...
 $ V79YTHE : num  0 0 0 0 0 0 0 0 0 0 ...
 $ V80RF   : num  0 0 0 0 0 1.4 0 1.9 3.2 0 ...
 $ V81SHBC : num  0 0 1.4 0 0.7 0 0 2 1.3 0.6 ...
 $ V82AZKF : num  0 0 0 0 0 0 0 0.7 0 0 ...
 $ V83SFC  : num  0 0 0 0 0 3.4 0 1.2 1.9 1.2 ...
 $ V84YRTH : num  0 0 0 0 0 0 0 0 0 0 ...
 $ V85ROSE : num  0 0 0 0 0 0 0 0.6 0 0 ...
 $ V86BCOO : num  0 0 0 0 0 0 0 0 0.6 0 ...
 $ V87LFC  : num  0 0 1.8 0 0 0 2.4 0 0 0 ...
 $ V88WG   : num  0 0 0 0 0 0 0 0 0 0 ...
 $ V89PCOO : num  1.9 0 0 0 0 0 0 0 0 0 ...
 $ V90WTG  : num  0 0 0 0 0 0 0 0 0 0 ...
 $ V91NMIN : num  0.2 0 5.8 3.1 0 0 0 0 0 0 ...
 $ V92NFB  : num  0 0 0 0 0 0 0 0 0 0 ...
 $ V93DB   : num  0 0 0 0 0 0 0 0 0 0 ...
 $ V94RBEE : num  0 0 0 0 0 0 0 0 0 0 ...
 $ V95HBC  : num  0 0 0 0 0 0 0 0 0 0 ...
 $ V96DF   : num  0 0 0 0 0 0 0 0 0 0 ...
 $ V97PCL  : num  9.1 0 0 0 0 0 0 0 0 0 ...
  [list output truncated]
```

Check the habitat groups:

CODE

```r
table(Birds$HABITAT)
```

OUTPUT

```
Box-Ironbark Foothills Wo Gippsland Ma         Mixed Montane Fore River Red Gu
           4              4              4            17             4              4
```

Create the community matrix

For multivariate community analyses, we need only the species abundance columns.

The first two columns are site information, so the bird species matrix starts at column 3.

CODE

```r
BirdsData <- Birds[, 3:ncol(Birds)]

head(BirdsData[, 1:8])
```

OUTPUT

```
  V1GST V2EYR V3GF V4BTH V5GWH V6WTTR V7WEHE V8WNHE
1   3.4   0.0  0.0   0.0   0.0    0.0    0.0   11.9
2   3.4   9.2  0.0   0.0   0.0    0.0    0.0   11.5
3   8.4   3.8  0.7   2.8   0.0    0.0   10.7   12.3
4   3.0   5.0  0.0   5.0   2.0    0.0    3.0   10.0
5   5.6   5.6 12.9  12.2   9.5    2.1    7.9   28.6
6   8.1   4.1 10.9  24.5   5.6    6.7    9.4    6.7
```

CODE

```r
dim(BirdsData)
```

OUTPUT

```
[1]  37 102
```

Each row is a site.
Each column is a bird species.

Transform the data

Community data often contain many zeros and some very abundant species. If we analyse raw abundance data, the ordination may be dominated by only the most abundant species.

A **fourth-root transformation** reduces the influence of very abundant species while still retaining quantitative abundance information.

```
CODE
TransBirdsData <- BirdsData^(1/4)
```

Bray-Curtis dissimilarity

Bray-Curtis dissimilarity is commonly used for ecological community data.

It compares sites based on differences in species composition and abundance.

```
CODE
Bird.dis <- vegdist(TransBirdsData, method = "bray")

Bird.dis
```

```
OUTPUT
           1         2         3         4         5         6         7
2  0.5811093
3  0.5538884 0.3012592
4  0.5168095 0.3048933 0.2656237
5  0.7288140 0.4277846 0.3739545 0.4776578
6  0.7346606 0.4854075 0.4626322 0.4971884 0.2904080
7  0.6841188 0.4513940 0.3187727 0.3630946 0.3557227 0.3373155
8  0.7836038 0.5372931 0.5115252 0.5490394 0.3083633 0.2240857 0.4019478
9  0.7828807 0.5558152 0.5564663 0.5824275 0.3465332 0.2390345 0.4506996
10 0.7018559 0.5557679 0.4865599 0.5552278 0.3271147 0.3391277 0.4324359
11 0.8063526 0.6167779 0.5001552 0.6020000 0.3385922 0.3581620 0.4327462
12 0.7266312 0.5499390 0.4809957 0.5191379 0.3633039 0.3201244 0.4158595
13 0.6924704 0.5196598 0.4507475 0.4978200 0.3222033 0.2564058 0.3464840
14 0.8510533 0.6203627 0.5630760 0.6157355 0.3812674 0.3493667 0.5065226
15 0.7959633 0.6222646 0.5527953 0.5849477 0.4094507 0.3742980 0.5031913
16 0.8073635 0.5931764 0.5088703 0.5757578 0.3731528 0.3838064 0.4486967
17 0.6218105 0.5123680 0.4292987 0.5145471 0.4254575 0.4521966 0.4541890
18 0.6288168 0.5141771 0.4502084 0.5041746 0.3400412 0.3792721 0.3382631
19 0.7169316 0.4737855 0.4365329 0.5063463 0.2919399 0.3493836 0.3982871
20 0.6882119 0.6015094 0.5146904 0.5463852 0.3506307 0.3781261 0.4076429
21 0.7207458 0.4628075 0.4030716 0.4833199 0.2213818 0.2272425 0.3184691
22 0.7130669 0.3963048 0.4435605 0.4157735 0.2887657 0.3190927 0.4548913
23 0.7554177 0.4648294 0.4673510 0.4573979 0.3576220 0.3357695 0.3699074
24 0.6578409 0.3261971 0.2845718 0.3033146 0.3974232 0.4653503 0.3550889
25 0.7564742 0.6557555 0.5304423 0.6354600 0.3886521 0.4254396 0.4272677
26 0.8274156 0.7185738 0.6055273 0.6441336 0.4908169 0.4240129 0.5252044
27 0.7135837 0.5642955 0.4683315 0.5301566 0.3160961 0.3563647 0.3996958
28 0.7518473 0.5009149 0.4421934 0.5193656 0.2893234 0.2855187 0.3778628
29 0.4531177 0.6256985 0.5994287 0.5463836 0.7257647 0.7478554 0.6379375
30 0.5565948 0.6965908 0.6312407 0.6125685 0.6996847 0.6982420 0.5969278
31 0.4446357 0.6983750 0.5853321 0.6002661 0.7165809 0.7143934 0.6339431
32 0.4199185 0.6335318 0.5783585 0.4944642 0.7458877 0.7876667 0.6852894
33 0.6319161 0.6323419 0.5275781 0.5963714 0.4590329 0.4614880 0.5310118
34 0.6294810 0.6351601 0.5177020 0.5992538 0.3705688 0.4405296 0.5484207
35 0.3936613 0.5282705 0.4333633 0.4680113 0.5600997 0.6449680 0.5806360
36 0.5940169 0.6182414 0.4841367 0.5814368 0.3857289 0.4355690 0.4637640
37 0.6700454 0.5603731 0.4565535 0.5501178 0.3244759 0.3876113 0.4089532
           8         9        10        11        12        13        14
2
```

```
3
4
5
6
7
8
9  0.1951151
10 0.3729687 0.3451244
11 0.2717601 0.3015726 0.2454991
12 0.2758298 0.2205846 0.3782381 0.2974830
13 0.2223679 0.2226500 0.3022838 0.2391993 0.2064072
14 0.2787909 0.2106467 0.4410868 0.3083956 0.2371092 0.3134122
15 0.2957034 0.2363472 0.4193424 0.3057258 0.1709806 0.2850145 0.1613859
16 0.2792215 0.2241250 0.4070212 0.2869727 0.2355982 0.2574916 0.2239040
17 0.4841469 0.4539468 0.3891711 0.4753925 0.4742680 0.4427194 0.5274046
18 0.3698541 0.4256049 0.4199316 0.3479013 0.3665011 0.3197675 0.4132451
19 0.3448985 0.3535099 0.2735383 0.2536778 0.3190726 0.2979499 0.3937406
20 0.4032488 0.4337080 0.2655589 0.2977125 0.3582272 0.2861492 0.4448466
21 0.2674204 0.2833619 0.2760638 0.3092336 0.3071221 0.2366256 0.3122786
22 0.3252211 0.3527112 0.4275477 0.4272384 0.4145930 0.3576349 0.3963489
23 0.4007785 0.3676908 0.3715669 0.4481153 0.4011944 0.3888619 0.4561900
24 0.5083378 0.5298122 0.5301139 0.5018120 0.4251041 0.3929244 0.5399769
25 0.4648920 0.5136172 0.3287552 0.4046693 0.4807776 0.4143108 0.5661840
26 0.3615145 0.3042304 0.4915418 0.4028192 0.2625627 0.3408723 0.2584830
27 0.2781346 0.2495484 0.3170386 0.2401769 0.2479473 0.2026242 0.3126309
28 0.2712258 0.2955475 0.3669449 0.3116880 0.2560819 0.2393910 0.3028019
29 0.7759636 0.7209046 0.6281837 0.6958882 0.6936614 0.6787960 0.7493157
30 0.7159321 0.7177126 0.5593490 0.6924704 0.7215431 0.6940181 0.7843145
31 0.7356662 0.7161492 0.5865631 0.6970164 0.7090184 0.6692351 0.7842994
32 0.8027009 0.8004714 0.7001694 0.7790406 0.7644425 0.7485170 0.8350025
33 0.4901806 0.5099387 0.3462249 0.4307006 0.4683899 0.3941127 0.6080585
34 0.4926852 0.4841842 0.3194907 0.4087302 0.4634862 0.4254745 0.5430026
35 0.6613886 0.6645110 0.5425301 0.6348805 0.5855076 0.5589297 0.7292768
36 0.4381436 0.4529427 0.2946949 0.3775072 0.3930000 0.3531846 0.5209080
37 0.3567463 0.4113019 0.2575492 0.3221561 0.3205038 0.2988627 0.4585136
          15        16        17        18        19        20        21
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16 0.1895676
17 0.5117597 0.4682845
18 0.3823999 0.3481498 0.3154883
19 0.3517541 0.3317676 0.4815143 0.3535642
20 0.3778650 0.4184401 0.4768728 0.3427651 0.2776771
21 0.3196154 0.3101225 0.4230529 0.3713238 0.2762805 0.3394350
22 0.3948861 0.3740458 0.4790184 0.4235341 0.3677298 0.4639006 0.2553800
23 0.4500571 0.4381147 0.4205407 0.4576355 0.4008931 0.4771469 0.2552954
24 0.4904771 0.4784935 0.4323657 0.3680164 0.4560534 0.4966671 0.4067531
25 0.5251346 0.5025636 0.5191401 0.4863091 0.3852502 0.3700371 0.3425710
26 0.2428673 0.2709968 0.5601140 0.4732700 0.4319630 0.4850794 0.4318570
27 0.2487586 0.1879095 0.4394022 0.3045709 0.2766347 0.3170154 0.2977140
28 0.2851350 0.3187882 0.4377821 0.3762856 0.3792002 0.3476928 0.1653544
29 0.7304746 0.7216215 0.4706448 0.5770535 0.6891767 0.6522839 0.7263503
30 0.7711130 0.7399037 0.4099727 0.5995901 0.6740752 0.6443037 0.6454932
```

```
31 0.7697285 0.7376919 0.4553512 0.5903320 0.6651305 0.6405915 0.6875499
32 0.8007412 0.7928213 0.4998473 0.6263614 0.7416589 0.7330399 0.7707607
33 0.5159623 0.5118776 0.4950797 0.5026771 0.4331789 0.3777457 0.4190648
34 0.4991433 0.5215556 0.5160834 0.5069106 0.3507247 0.3386141 0.3738222
35 0.6424039 0.6436653 0.4773244 0.5105333 0.5668449 0.5273110 0.5959499
36 0.4753619 0.4808798 0.4595026 0.4644444 0.3737548 0.3067652 0.3430217
37 0.4091016 0.4238754 0.4456725 0.3813058 0.2548519 0.2078700 0.2918365
          22        23        24        25        26        27        28
2
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19
20
21
22
23 0.2931821
24 0.4083533 0.3880791
25 0.5574525 0.4645878 0.5039462
26 0.5364468 0.5802964 0.5741870 0.6119036
27 0.3812239 0.4063233 0.4310905 0.4381482 0.3262531
28 0.2825661 0.3212464 0.4186415 0.4277676 0.4456717 0.3102918
29 0.7579502 0.6825325 0.5779074 0.7216651 0.7289317 0.6850755 0.7333197
30 0.7669952 0.6707639 0.6821498 0.6050880 0.7661826 0.7008594 0.7055185
31 0.7772150 0.7069920 0.7177544 0.6333638 0.7337614 0.6958954 0.7424756
32 0.7820377 0.7662020 0.6293694 0.7544690 0.8168436 0.7574892 0.7990645
33 0.5408097 0.5194035 0.4946126 0.2631791 0.6205043 0.4609607 0.4206607
34 0.5204719 0.4994480 0.5324876 0.2478243 0.5855205 0.4575362 0.4133860
35 0.6017573 0.6591710 0.4996661 0.5393154 0.7531367 0.5558782 0.5947992
36 0.5108520 0.4842274 0.4955954 0.2930084 0.5419917 0.4133552 0.3836528
37 0.4343973 0.4248411 0.4837006 0.3358708 0.4954951 0.3376827 0.3357712
          29        30        31        32        33        34        35
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30 0.3957464
31 0.3082678 0.2172576
32 0.2121637 0.3805270 0.3036573
33 0.6752249 0.5995809 0.6008674 0.6838187
34 0.6768839 0.6314822 0.6503030 0.7069511 0.1945662
35 0.4733505 0.5019833 0.4990476 0.4264275 0.4308685 0.4676846
36 0.6242101 0.5747802 0.6046206 0.6708376 0.2282433 0.2260476 0.3915569
37 0.6887188 0.6069576 0.6060879 0.7142322 0.3269716 0.3210003 0.4916419
          36
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25
26
27
28
29
30
31
32
33
34
35
36
37 0.2166400
```

Interpretation:

- values close to 0 = sites have similar bird communities
- values close to 1 = sites have very different bird communities

nMDS ordination

nMDS attempts to place sites in a low-dimensional plot so that distances among points reflect their Bray-Curtis dissimilarities.

Sites close together have similar bird assemblages.
Sites far apart have different bird assemblages.

CODE

```
set.seed(123)
nMDS_Bird <- metaMDS(TransBirdsData,
                     distance = "bray",
                     k = 2,
                     trymax = 100)
```

OUTPUT

```
Run 0 stress 0.1112209
Run 1 stress 0.1181205
Run 2 stress 0.1268505
Run 3 stress 0.1112209
... New best solution
... Procrustes: rmse 1.04629e-06  max resid 4.436971e-06
... Similar to previous best
Run 4 stress 0.1112209
... Procrustes: rmse 5.289056e-06  max resid 1.95247e-05
... Similar to previous best
Run 5 stress 0.1181205
Run 6 stress 0.1145319
Run 7 stress 0.111221
... Procrustes: rmse 5.574731e-05  max resid 0.0002576911
... Similar to previous best
Run 8 stress 0.1145319
Run 9 stress 0.111221
... Procrustes: rmse 9.96313e-05  max resid 0.0004700412
... Similar to previous best
Run 10 stress 0.1112209
... Procrustes: rmse 2.47593e-05  max resid 0.0001153228
... Similar to previous best
Run 11 stress 0.1298771
Run 12 stress 0.1112209
... New best solution
... Procrustes: rmse 2.937168e-06  max resid 1.411304e-05
... Similar to previous best
Run 13 stress 0.1181205
Run 14 stress 0.1112209
... New best solution
... Procrustes: rmse 2.35266e-06  max resid 1.158071e-05
... Similar to previous best
Run 15 stress 0.1112209
... Procrustes: rmse 1.305642e-05  max resid 6.065315e-05
... Similar to previous best
Run 16 stress 0.1145319
Run 17 stress 0.1181205
Run 18 stress 0.1181205
Run 19 stress 0.1268648
Run 20 stress 0.1181205
*** Best solution repeated 2 times
```

CODE

```r
nMDS_Bird
```

OUTPUT

```
Call:
metaMDS(comm = TransBirdsData, distance = "bray", k = 2, trymax = 100)

global Multidimensional Scaling using monoMDS

Data:     TransBirdsData
Distance: bray

Dimensions: 2
Stress:     0.1112209
Stress type 1, weak ties
Best solution was repeated 2 times in 20 tries
The best solution was from try 14 (random start)
Scaling: centring, PC rotation, halfchange scaling
Species: expanded scores based on 'TransBirdsData'
```

Stress value

The stress value tells us how well the two-dimensional nMDS represents the multivariate dissimilarities.

As a guide:

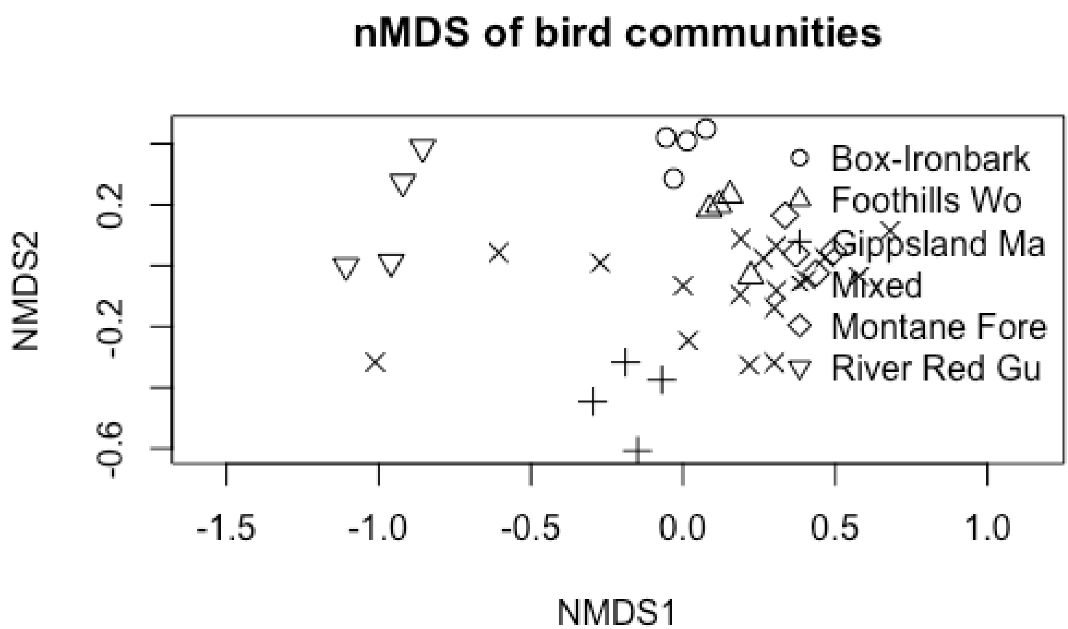
- stress < 0.10 = good representation
- stress 0.10–0.20 = usable but interpret with care
- stress > 0.20 = poor representation

Base R nMDS plot

CODE

```r
Bird_Factor <- factor(Birds$HABITAT)
pchs <- seq_along(levels(Bird_Factor))

plot(nMDS_Bird,
     display = "sites",
     type = "n",
     main = "nMDS of bird communities")

points(nMDS_Bird,
       display = "sites",
       pch = pchs[Bird_Factor],
       col = "black",
       cex = 1.2)
```

```
legend("topright",
       bty = "n",
       legend = levels(Bird_Factor),
       pch = pchs)
```



Interpretation

If sites from the same habitat type cluster together, this suggests that habitat type is associated with bird community composition.

If habitat groups strongly overlap, this suggests that bird communities are not clearly separated by habitat type.

ggplot nMDS plot

First extract the nMDS site scores.

CODE

```r
nmds_scores <- as.data.frame(scores(nMDS_Bird, display = "sites"))
nmds_scores$SITE <- Birds$SITE
nmds_scores$HABITAT <- Birds$HABITAT

head(nmds_scores)
```

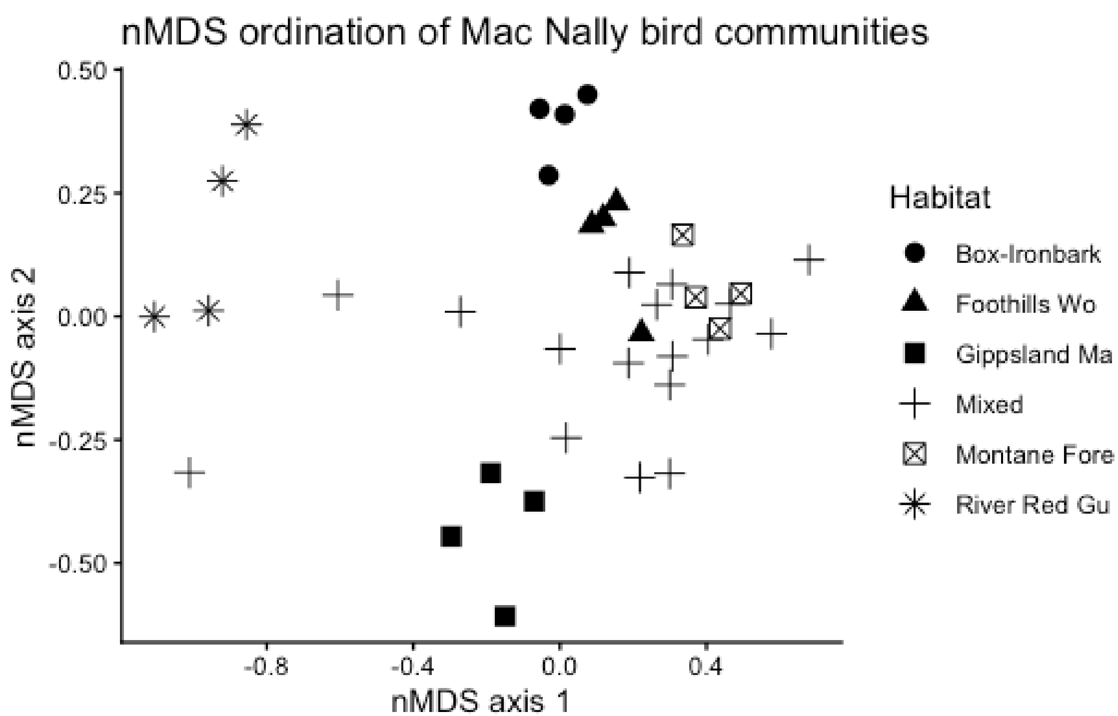
OUTPUT

```
       NMDS1       NMDS2        SITE      HABITAT
1 -1.0111806 -0.31642471  Reedy Lake        Mixed
2 -0.1497519 -0.60758304  Pearcedale Gippsland Ma
3 -0.1887677 -0.31707278     Warneet Gippsland Ma
4 -0.2965144 -0.44565339  Cranbourne Gippsland Ma
5  0.1872838 -0.09485409 Lysterfield        Mixed
6  0.2999934 -0.13865712    Red Hill        Mixed
```

Now plot the ordination.

CODE

```r
ggplot(nmds_scores, aes(x = NMDS1, y = NMDS2, shape = HABITAT)) +
  geom_point(size = 3) +
  theme_classic() +
  labs(title = "nMDS ordination of Mac Nally bird communities",
       x = "nMDS axis 1",
       y = "nMDS axis 2",
       shape = "Habitat")
```

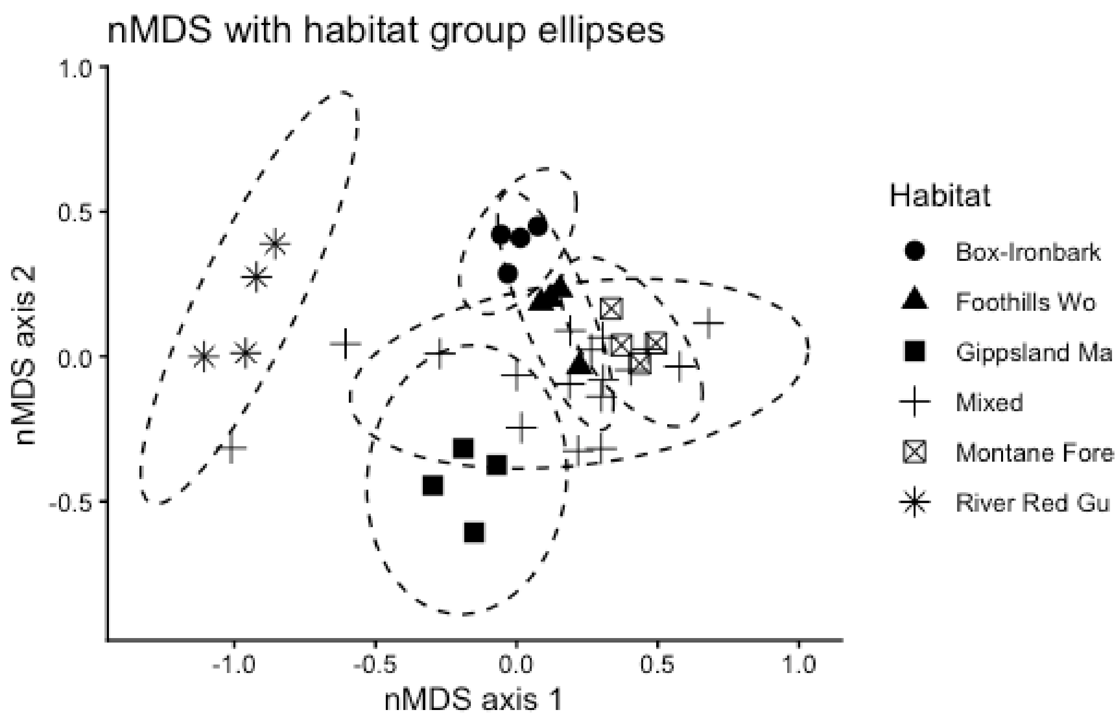


Add habitat ellipses

Ellipses help visualise whether habitat groups occupy different regions of ordination space.

CODE

```
ggplot(nmds_scores, aes(x = NMDS1, y = NMDS2, shape = HABITAT)) +
  geom_point(size = 3) +
  stat_ellipse(aes(group = HABITAT), type = "t", linetype = 2) +
  theme_classic() +
  labs(title = "nMDS with habitat group ellipses",
       x = "nMDS axis 1",
       y = "nMDS axis 2",
       shape = "Habitat")
```



Note: ellipses are only a visual guide. Formal tests are below.

ANOSIM

ANOSIM tests whether differences among habitat groups are greater than differences within habitat groups.

CODE

```r
Bird.anosim <- anosim(Bird.dis, Birds$HABITAT, permutations = 999)

summary(Bird.anosim)
```

OUTPUT

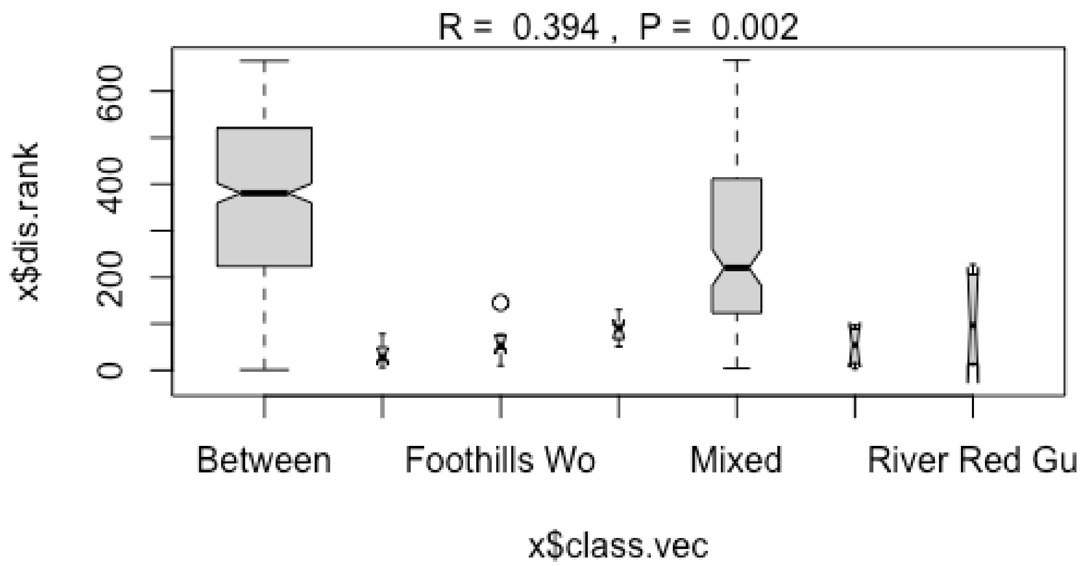
```

Call:
anosim(x = Bird.dis, grouping = Birds$HABITAT, permutations = 999)
Dissimilarity: bray

ANOSIM statistic R: 0.3936
      Significance: 0.002

Permutation: free
Number of permutations: 999

Upper quantiles of permutations (null model):
  90%   95% 97.5%   99%
0.127 0.171 0.202 0.249

Dissimilarity ranks between and within classes:
             0%    25%   50%    75% 100%   N
Between       1 224.25 380.5 520.25  665 500
Box-Ironbark  6  22.50  29.5  45.50   78   6
Foothills Wo 10  47.00  54.0  71.50  145   6
Gippsland Ma 51  72.25  90.0  95.00  130   6
Mixed         4 124.00 220.5 410.25  666 136
Montane Fore  3  17.75  54.5  87.50   97   6
River Red Gu 12  33.75  96.5 179.50  227   6
```

CODE

```r
plot(Bird.anosim)
```

OUTPUT

```
Warning in (function (z, notch = FALSE, width = NULL, varwidth = FALSE, : some
notches went outside hinges ('box'): maybe set notch=FALSE
```



Interpreting ANOSIM

The main statistic is **R**.

- R close to 0 = little separation among groups
- R close to 1 = strong separation among groups
- p-value = whether the observed separation is greater than expected by chance

PERMANOVA

PERMANOVA tests whether habitat type explains variation in the multivariate bird community data.

In `vegan`, PERMANOVA is run using `adonis2()`.

```
CODE
Bird.permanova <- adonis2(TransBirdsData ~ HABITAT,
                          data = Birds,
                          method = "bray",
```

```
                              permutations = 999)

Bird.permanova
```

```
OUTPUT
Permutation test for adonis under reduced model
Permutation: free
Number of permutations: 999

adonis2(formula = TransBirdsData ~ HABITAT, data = Birds, permutations = 999, method = "bray")
         Df SumOfSqs      R2      F Pr(>F)
Model     5   2.2749 0.50098 6.2245  0.001 ***
Residual 31   2.2660 0.49902
Total    36   4.5409 1.00000
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Interpreting PERMANOVA

Important columns:

- `Df` = degrees of freedom
- `SumOfSqs` = multivariate variation explained
- `R2` = proportion of variation explained by habitat
- `F` = pseudo-F statistic
- `Pr(>F)` = permutation p-value

If the p-value is significant, bird communities differ among habitat types.

Check dispersion

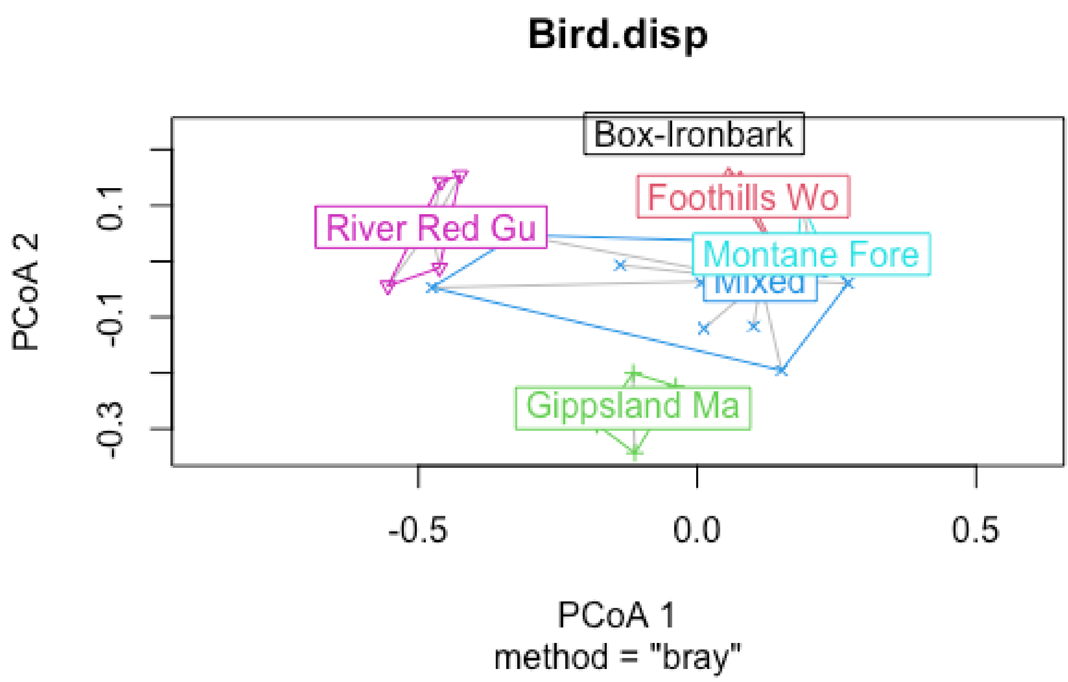
PERMANOVA can be sensitive to differences in multivariate spread among groups. Therefore, we should check whether habitat groups differ in dispersion.

```
CODE
Bird.disp <- betadisper(Bird.dis, Birds$HABITAT)

anova(Bird.disp)
```

```
OUTPUT
Analysis of Variance Table

Response: Distances
          Df  Sum Sq   Mean Sq F value  Pr(>F)
Groups     5 0.14637 0.0292742  3.0384 0.02403 *
Residuals 31 0.29868 0.0096347
```

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
CODE
plot(Bird.disp)
```



Interpretation

If this test is significant, habitat groups differ in their within-group variability.

That does not invalidate PERMANOVA, but it means the PERMANOVA result may reflect differences in spread as well as differences in group centroids.

Pairwise PERMANOVA comparisons

If the overall PERMANOVA is significant, we may want to know which habitat groups differ from each other.

The package `pairwiseAdonis` is useful, but it may not be installed on all computers. The installation code is shown but not run automatically.

```
CODE
# Install only once, if needed:
# install.packages("devtools")
# devtools::install_github("pmartinezarbizu/pairwiseAdonis/pairwiseAdonis")
```

Load the package if it is already installed:

```
CODE
library(pairwiseAdonis)

pairwise.adonis(TransBirdsData,
                Birds$HABITAT,
                sim.method = "bray",
                p.adjust.m = "bonferroni")
```

If this package is not available, you can still interpret the overall PERMANOVA and the nMDS plot.

SIMPER analysis

SIMPER stands for **Similarity Percentages**.

It identifies which species contribute most to differences between habitat groups.

```
CODE
Birdsim <- simper(TransBirdsData, Birds$HABITAT)

summary(Birdsim)
```

```
OUTPUT

Contrast: Mixed_Gippsland Ma

          average       sd    ratio      ava      avb cumsum     p
V17WPHE  0.019779 0.011557 1.711400 0.602000 2.247700  0.040 0.003 **
V34WSW   0.018979 0.007112 2.668500 0.247500 1.856000  0.079 0.001 ***
V11CR    0.016850 0.007496 2.247800 1.443400 0.000000  0.114 0.001 ***
V6WTTR   0.015456 0.006359 2.430500 1.325800 0.000000  0.145 0.002 **
V15STTH  0.014940 0.009012 1.657800 1.475400 0.323400  0.176 0.008 **
V25WAG   0.013966 0.005802 2.407300 0.205100 1.376000  0.204 0.002 **
V27NHHE  0.012769 0.008435 1.513800 0.689200 1.646300  0.231 0.044 *
V3GF     0.012483 0.008488 1.470700 1.569300 0.646500  0.256 0.001 ***
V4BTH    0.011472 0.009944 1.153700 1.946400 1.184200  0.280 0.129
V36YFHE  0.010708 0.008006 1.337500 1.121400 1.243700  0.302 0.086 .
V42SIL   0.010651 0.008608 1.237400 0.952800 0.420400  0.323 0.029 *
V13RWB   0.010585 0.007337 1.442700 1.080500 1.922100  0.345 0.156
```

```
V33RWH   0.010302 0.006859 1.501900 1.318700 0.579800  0.366 0.031 *
V5GWH    0.010203 0.008334 1.224100 1.325200 0.674800  0.387 0.006 **
V16LR    0.009557 0.008123 1.176500 0.852900 1.184900  0.407 0.102
V65LWB   0.009521 0.009391 1.013900 0.258700 0.741500  0.426 0.053 .
V21ESP   0.009484 0.007739 1.225400 1.236300 0.644000  0.445 0.130
V23RBFT  0.009142 0.006803 1.343900 0.990600 0.937700  0.464 0.275
V63DWS   0.008333 0.007483 1.113600 0.313100 0.724800  0.481 0.020 *
V35STP   0.008318 0.007536 1.103700 0.700100 0.993200  0.498 0.338
V7WEHE   0.008195 0.007490 1.094100 1.252500 1.241500  0.515 0.162
V91NMIN  0.008158 0.007918 1.030300 0.103300 0.719700  0.532 0.033 *
V19ER    0.007905 0.006671 1.184900 0.816000 1.308600  0.548 0.355
V29CST   0.007715 0.005456 1.414100 0.548200 1.059100  0.564 0.079 .
V56FTC   0.007610 0.006779 1.122500 0.948600 0.615700  0.579 0.106
V14AUR   0.007440 0.007024 1.059200 0.893600 0.962100  0.594 0.176
V81SHBC  0.007124 0.006361 1.120000 0.641600 0.271900  0.609 0.107
V28VS    0.007070 0.008926 0.792000 0.599300 0.000000  0.624 0.900
V12LK    0.006733 0.005709 1.179300 1.372500 0.801000  0.637 0.009 **
V80RF    0.006689 0.007273 0.919800 0.568000 0.000000  0.651 0.251
V50CHE   0.006645 0.008280 0.802400 0.580200 0.000000  0.664 0.318
V46BHHE  0.006634 0.006911 0.959900 0.491200 0.307900  0.678 0.944
V9SFW    0.006024 0.006604 0.912200 1.384400 1.734700  0.690 0.391
V48YTBC  0.005842 0.007167 0.815200 0.511500 0.000000  0.702 0.378
V41SPP   0.005728 0.005620 1.019200 1.058800 1.200500  0.714 0.702
V8WNHE   0.005294 0.006131 0.863400 1.588600 1.843200  0.725 0.844
V24BFCS  0.005165 0.005210 0.991300 0.751600 1.112500  0.736 0.545
V37WHIP  0.005130 0.007087 0.723800 0.441900 0.000000  0.746 0.490
V83SFC   0.005026 0.007077 0.710200 0.424600 0.000000  0.756 0.680
V43GCU   0.004955 0.006908 0.717300 0.423100 0.000000  0.766 0.939
V32SCC   0.004898 0.010846 0.451600 0.441300 0.000000  0.776 0.817
V20PCU   0.004806 0.006607 0.727400 0.411800 0.000000  0.786 0.948
V38GAL   0.004767 0.007107 0.670700 0.178700 0.323400  0.796 0.581
V87LFC   0.004719 0.006574 0.717800 0.224900 0.289600  0.806 0.134
V10WBSW  0.004538 0.005738 0.791000 1.426000 1.760400  0.815 1.000
V2EYR    0.004523 0.005813 0.778100 1.285300 1.599500  0.824 0.895
V53MB    0.004509 0.006393 0.705200 0.224200 0.250000  0.833 0.423
V31AMAG  0.004040 0.007500 0.538600 0.370900 0.000000  0.842 0.771
V55LHE   0.004024 0.006359 0.632900 0.354900 0.000000  0.850 0.482
V45MGLK  0.003945 0.005906 0.668000 0.142600 0.271900  0.858 0.522
V69SKF   0.003692 0.005819 0.634500 0.334500 0.000000  0.866 0.942
V70RSL   0.003469 0.005750 0.603200 0.094600 0.236400  0.873 0.703
V49LYRE  0.003229 0.005942 0.543500 0.268000 0.000000  0.879 0.646
V85ROSE  0.002816 0.005168 0.545000 0.239700 0.000000  0.885 0.463
V39FHE   0.002732 0.007692 0.355200 0.248500 0.000000  0.891 0.821
V68PIL   0.002684 0.005897 0.455200 0.232900 0.000000  0.896 0.671
V44MUSK  0.002563 0.007101 0.361000 0.236300 0.000000  0.901 0.730
V98FLAME 0.002557 0.005598 0.456700 0.220500 0.000000  0.907 0.723
V51OWH   0.002408 0.005315 0.453100 0.211500 0.000000  0.911 0.405
V1GST    0.002323 0.001704 1.363300 1.466600 1.493300  0.916 0.309
V22SCR   0.002318 0.005111 0.453500 0.210400 0.000000  0.921 0.999
V52TRM   0.002247 0.006427 0.349700 0.202800 0.000000  0.926 0.714
V47RFC   0.002229 0.004899 0.455000 0.214300 0.000000  0.930 0.891
V30BTR   0.002214 0.006274 0.352900 0.200700 0.000000  0.935 0.943
V58OBO   0.002124 0.004652 0.456500 0.198800 0.000000  0.939 0.984
V26WWCH  0.002007 0.005550 0.361500 0.180300 0.000000  0.943 1.000
V77RRP   0.001900 0.005290 0.359200 0.174100 0.000000  0.947 0.223
V76GBB   0.001874 0.004111 0.455800 0.166700 0.000000  0.951 0.548
V18YTH   0.001726 0.004872 0.354300 0.171800 0.000000  0.954 1.000
V86BCOO  0.001595 0.004421 0.360800 0.142300 0.000000  0.958 0.573
V67GGC   0.001558 0.004376 0.356000 0.135400 0.000000  0.961 0.787
V64BELL  0.001553 0.006272 0.247500 0.127500 0.000000  0.964 0.409
V62RBTR  0.001522 0.004260 0.357300 0.127400 0.000000  0.967 0.601
V57PINK  0.001480 0.004113 0.359800 0.127300 0.000000  0.970 0.376
V92NFB   0.001412 0.003904 0.361800 0.141400 0.000000  0.973 0.592
V97PCL   0.001237 0.004997 0.247500 0.102200 0.000000  0.976 0.118
V40BRTH  0.001151 0.004652 0.247400 0.088800 0.000000  0.978 0.989
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V78LLOR  0.001038 0.004191 0.247800 0.106900 0.000000  0.980 0.120
V60LFB   0.000929 0.003754 0.247500 0.076800 0.000000  0.982 0.961
V84YRTH  0.000851 0.003434 0.247700 0.082700 0.000000  0.984 0.621
V89PCOO  0.000836 0.003378 0.247500 0.069100 0.000000  0.985 0.118
V54STHR  0.000766 0.003093 0.247700 0.073200 0.000000  0.987 0.847
V66CBW   0.000735 0.002968 0.247500 0.058800 0.000000  0.989 0.864
V79YTHE  0.000733 0.002957 0.247800 0.075400 0.000000  0.990 0.585
V96DF    0.000688 0.002777 0.247800 0.070800 0.000000  0.991 0.739
V82AZKF  0.000680 0.002747 0.247500 0.053800 0.000000  0.993 0.114
V100WBWS 0.000643 0.002594 0.247800 0.066200 0.000000  0.994 0.120
V93DB    0.000634 0.002557 0.247700 0.061600 0.000000  0.995 0.942
V102KING 0.000608 0.002454 0.247600 0.053800 0.000000  0.997 0.428
V90WTG   0.000572 0.002311 0.247700 0.055600 0.000000  0.998 0.958
V71PDOV  0.000540 0.002181 0.247800 0.055600 0.000000  0.999 0.786
V94RBEE  0.000523 0.002110 0.247800 0.053800 0.000000  1.000 0.981
V59YR    0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V61SPW   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V72CRP   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V73JW    0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V74BCHE  0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V75RCR   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V88WG    0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V95HBC   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V99WWT   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V101LCOR 0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
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Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Contrast: Mixed_Montane Fore

          average       sd    ratio      ava      avb cumsum     p
V13RWB   0.009338 0.006981 1.337700 1.080500 0.662400  0.025 0.525
V43GCU   0.009042 0.005499 1.644400 0.423100 1.256100  0.049 0.011 *
V83SFC   0.008981 0.004949 1.814500 0.424600 1.220000  0.073 0.016 *
V20PCU   0.008970 0.005919 1.515400 0.411800 1.268100  0.097 0.015 *
V67GGC   0.008783 0.005683 1.545500 0.135400 0.926600  0.121 0.003 **
V16LR    0.008555 0.006520 1.312100 0.852900 1.023200  0.144 0.335
V46BHHE  0.008352 0.007086 1.178600 0.491200 0.828800  0.166 0.535
V33RWH   0.008286 0.006874 1.205500 1.318700 0.753700  0.188 0.224
V37WHIP  0.008217 0.006567 1.251200 0.441900 1.020400  0.211 0.026 *
V23RBFT  0.008056 0.006201 1.299200 0.990600 0.883700  0.232 0.712
V36YFHE  0.008026 0.007167 1.120000 1.121400 1.134500  0.254 0.883
V28VS    0.007929 0.007261 1.091900 0.599300 0.709200  0.275 0.616
V19ER    0.007909 0.006742 1.173100 0.816000 0.266900  0.296 0.390
V48YTBC  0.007759 0.006195 1.252500 0.511500 0.958400  0.317 0.038 *
V54STHR  0.007620 0.004661 1.634900 0.073200 0.758900  0.337 0.003 **
V68PIL   0.007514 0.004974 1.510900 0.232900 0.785200  0.358 0.007 **
V35STP   0.007322 0.006426 1.139400 0.700100 0.944200  0.377 0.787
V27NHHE  0.007298 0.008654 0.843300 0.689200 0.000000  0.397 0.692
V49LYRE  0.007170 0.005092 1.408100 0.268000 0.803100  0.416 0.009 **
V50CHE   0.006930 0.005950 1.164700 0.580200 0.509600  0.435 0.263
V80RF    0.006902 0.006031 1.144500 0.568000 0.913500  0.453 0.159
V42SIL   0.006868 0.005994 1.145900 0.952800 1.363600  0.472 0.960
V14AUR   0.006236 0.005868 1.062700 0.893600 0.919800  0.488 0.423
V55LHE   0.006136 0.005947 1.031900 0.354900 0.590500  0.505 0.094 .
V29CST   0.006120 0.006185 0.989500 0.548200 0.314400  0.521 0.873
V24BFCS  0.006092 0.005192 1.173200 0.751600 0.496700  0.537 0.229
V26WWCH  0.006052 0.009091 0.665800 0.180300 0.462400  0.554 0.978
V17WPHE  0.005965 0.009471 0.629800 0.602000 0.000000  0.570 0.991
V81SHBC  0.005961 0.005479 1.088100 0.641600 1.191300  0.586 0.867
V8WNHE   0.005525 0.005093 1.084800 1.588600 1.709100  0.601 0.795
V21ESP   0.005370 0.004726 1.136100 1.236300 1.502100  0.615 0.999
V15STTH  0.005290 0.006652 0.795300 1.475400 1.857500  0.629 0.956
V86BCOO  0.005217 0.005078 1.027400 0.142300 0.481700  0.643 0.038 *
V11CR    0.004834 0.005601 0.863000 1.443400 1.710400  0.656 1.000
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V41SPP   0.004678 0.005784 0.808700 1.058800 1.352200  0.669 0.889
V22SCR   0.004598 0.006404 0.718100 0.210400 0.317500  0.681 0.913
V85ROSE  0.004568 0.005803 0.787200 0.239700 0.331700  0.693 0.086 .
V9SFW    0.004450 0.004930 0.902600 1.384400 1.525000  0.705 0.794
V32SCC   0.004345 0.009627 0.451300 0.441300 0.000000  0.717 0.880
V4BTH    0.004124 0.005531 0.745600 1.946400 2.294000  0.728 0.996
V56FTC   0.004054 0.005016 0.808300 0.948600 1.231300  0.739 0.992
V6WTTR   0.003848 0.005024 0.765900 1.325800 1.567100  0.749 0.982
V62RBTR  0.003800 0.005744 0.661600 0.127400 0.281200  0.759 0.115
V7WEHE   0.003646 0.003640 1.001600 1.252500 1.224500  0.769 0.993
V31AMAG  0.003590 0.006655 0.539500 0.370900 0.000000  0.779 0.852
V51OWH   0.003548 0.004941 0.718000 0.211500 0.220000  0.788 0.225
V5GWH    0.003405 0.004259 0.799600 1.325200 1.486100  0.797 0.962
V2EYR    0.003300 0.004546 0.725800 1.285300 1.440100  0.806 0.991
V69SKF   0.003277 0.005158 0.635200 0.334500 0.000000  0.815 0.983
V57PINK  0.003208 0.004800 0.668200 0.127300 0.250000  0.824 0.149
V76GBB   0.003203 0.004482 0.714700 0.166700 0.220000  0.832 0.173
V63DWS   0.003119 0.004987 0.625500 0.313100 0.000000  0.840 0.850
V10WBSW  0.002993 0.004868 0.614800 1.426000 1.622600  0.849 1.000
V3GF     0.002942 0.004175 0.704800 1.569300 1.724600  0.856 0.996
V65LWB   0.002877 0.008006 0.359400 0.258700 0.000000  0.864 0.619
V102KING 0.002549 0.004128 0.617500 0.053800 0.236400  0.871 0.102
V39FHE   0.002424 0.006810 0.356000 0.248500 0.000000  0.877 0.888
V34WSW   0.002385 0.005233 0.455700 0.247500 0.000000  0.884 0.989
V53MB    0.002366 0.005345 0.442500 0.224200 0.000000  0.890 0.894
V44MUSK  0.002278 0.006306 0.361200 0.236300 0.000000  0.896 0.800
V98FLAME 0.002256 0.004937 0.457000 0.220500 0.000000  0.902 0.802
V87LFC   0.002212 0.004906 0.450900 0.224900 0.000000  0.908 0.700
V52TRM   0.001993 0.005682 0.350700 0.202800 0.000000  0.914 0.811
V47RFC   0.001992 0.004375 0.455300 0.214300 0.000000  0.919 0.936
V30BTR   0.001964 0.005551 0.353800 0.200700 0.000000  0.924 0.956
V25WAG   0.001953 0.004286 0.455700 0.205100 0.000000  0.929 0.998
V58OBO   0.001891 0.004139 0.456800 0.198800 0.000000  0.934 0.989
V1GST    0.001754 0.001220 1.437400 1.466600 1.379200  0.939 0.857
V38GAL   0.001728 0.004792 0.360600 0.178700 0.000000  0.944 0.977
V77RRP   0.001687 0.004691 0.359700 0.174100 0.000000  0.948 0.414
V18YTH   0.001548 0.004368 0.354500 0.171800 0.000000  0.953 1.000
V45MGLK  0.001409 0.003896 0.361600 0.142600 0.000000  0.956 0.980
V64BELL  0.001363 0.005506 0.247500 0.127500 0.000000  0.960 0.542
V92NFB   0.001268 0.003504 0.361700 0.141400 0.000000  0.963 0.806
V12LK    0.001220 0.000891 1.368900 1.372500 1.324100  0.967 0.998
V97PCL   0.001087 0.004390 0.247500 0.102200 0.000000  0.970 0.501
V70RSL   0.001006 0.004066 0.247500 0.094600 0.000000  0.972 0.989
V40BRTH  0.001003 0.004051 0.247500 0.088800 0.000000  0.975 0.993
V91NMIN  0.000978 0.002735 0.357500 0.103300 0.000000  0.978 0.973
V78LLOR  0.000935 0.003773 0.247700 0.106900 0.000000  0.980 0.538
V60LFB   0.000816 0.003298 0.247500 0.076800 0.000000  0.982 0.955
V84YRTH  0.000761 0.003073 0.247700 0.082700 0.000000  0.984 0.800
V89PCOO  0.000734 0.002967 0.247500 0.069100 0.000000  0.986 0.501
V79YTHE  0.000659 0.002662 0.247700 0.075400 0.000000  0.988 0.819
V66CBW   0.000643 0.002597 0.247500 0.058800 0.000000  0.990 0.927
V96DF    0.000619 0.002499 0.247700 0.070800 0.000000  0.991 0.828
V82AZKF  0.000594 0.002400 0.247500 0.053800 0.000000  0.993 0.515
V100WBWS 0.000579 0.002335 0.247700 0.066200 0.000000  0.995 0.538
V93DB    0.000567 0.002289 0.247700 0.061600 0.000000  0.996 0.970
V90WTG   0.000512 0.002068 0.247700 0.055600 0.000000  0.997 0.986
V71PDOV  0.000486 0.001964 0.247700 0.055600 0.000000  0.999 0.917
V94RBEE  0.000470 0.001899 0.247700 0.053800 0.000000  1.000 0.997
V59YR    0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V61SPW   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V72CRP   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V73JW    0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V74BCHE  0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V75RCR   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V88WG    0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
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V95HBC   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V99WWT   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V101LCOR 0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
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Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Contrast: Mixed_Foothills Wo

          average       sd    ratio      ava      avb cumsum     p
V40BRTH  0.014392 0.009230 1.559200 0.088800 1.423800  0.037 0.005 **
V22SCR   0.012437 0.004863 2.557800 0.210400 1.364500  0.068 0.002 **
V26WWCH  0.011907 0.007887 1.509700 0.180300 1.210300  0.099 0.020 *
V18YTH   0.011886 0.007877 1.508900 0.171800 1.199300  0.129 0.021 *
V10WBSW  0.009520 0.007570 1.257700 1.426000 0.773600  0.154 0.239
V23RBFT  0.009458 0.007442 1.270900 0.990600 0.243500  0.178 0.189
V28VS    0.009159 0.007107 1.288800 0.599300 1.088900  0.201 0.149
V53MB    0.008400 0.005483 1.532100 0.224200 0.851600  0.222 0.020 *
V46BHHE  0.008159 0.006034 1.352200 0.491200 0.931700  0.243 0.638
V16LR    0.008067 0.006758 1.193700 0.852900 1.479800  0.264 0.522
V19ER    0.007778 0.006808 1.142400 0.816000 0.654600  0.284 0.431
V69SKF   0.007708 0.006107 1.262200 0.334500 0.890600  0.304 0.050 *
V35STP   0.007702 0.006775 1.136900 0.700100 1.011200  0.323 0.677
V42SIL   0.007673 0.005328 1.440100 0.952800 0.761200  0.343 0.813
V27NHHE  0.007657 0.007842 0.976500 0.689200 0.256000  0.362 0.618
V50CHE   0.007225 0.007685 0.940200 0.580200 0.384600  0.381 0.256
V36YFHE  0.007150 0.005676 1.259600 1.121400 1.319700  0.399 0.967
V29CST   0.007090 0.006413 1.105700 0.548200 0.659900  0.417 0.272
V20PCU   0.007080 0.006583 1.075500 0.411800 0.673000  0.435 0.218
V9SFW    0.007048 0.006634 1.062400 1.384400 1.081900  0.453 0.219
V14AUR   0.006603 0.006170 1.070200 0.893600 0.987000  0.470 0.308
V43GCU   0.006573 0.006112 1.075300 0.423100 0.583000  0.487 0.494
V80RF    0.006348 0.006410 0.990200 0.568000 0.304500  0.503 0.399
V13RWB   0.006090 0.004647 1.310500 1.080500 1.170600  0.519 0.998
V17WPHE  0.006088 0.009642 0.631400 0.602000 0.000000  0.534 0.985
V64BELL  0.005858 0.009467 0.618800 0.127500 0.492000  0.549 0.134
V81SHBC  0.005798 0.005361 1.081400 0.641600 0.807100  0.564 0.928
V15STTH  0.005476 0.006465 0.847000 1.475400 1.788200  0.578 0.937
V83SFC   0.005382 0.006098 0.882700 0.424600 0.261700  0.592 0.581
V24BFCS  0.005280 0.004956 1.065500 0.751600 0.794400  0.605 0.495
V41SPP   0.005277 0.006727 0.784500 1.058800 1.526000  0.619 0.779
V48YTBC  0.005274 0.006448 0.817800 0.511500 0.000000  0.632 0.544
V8WNHE   0.005251 0.004942 1.062400 1.588600 1.582200  0.645 0.814
V11CR    0.004856 0.004784 1.015100 1.443400 1.484400  0.658 1.000
V21ESP   0.004753 0.004815 0.987100 1.236300 1.346100  0.670 1.000
V37WHIP  0.004623 0.006367 0.726100 0.441900 0.000000  0.682 0.665
V32SCC   0.004434 0.009801 0.452400 0.441300 0.000000  0.693 0.834
V63DWS   0.004331 0.005235 0.827300 0.313100 0.243500  0.704 0.552
V7WEHE   0.004274 0.004093 1.044200 1.252500 1.318200  0.715 0.951
V56FTC   0.004240 0.005159 0.821900 0.948600 1.244700  0.726 0.985
V33RWH   0.004130 0.005045 0.818600 1.318700 1.597500  0.736 0.985
V4BTH    0.004096 0.004957 0.826300 1.946400 1.975000  0.747 0.999
V96DF    0.003970 0.006435 0.617000 0.070800 0.349000  0.757 0.073 .
V58OBO   0.003798 0.005307 0.715600 0.198800 0.261700  0.767 0.794
V6WTTR   0.003785 0.005196 0.728400 1.325800 1.579900  0.776 0.970
V2EYR    0.003733 0.004281 0.872000 1.285300 1.316800  0.786 0.966
V31AMAG  0.003663 0.006775 0.540700 0.370900 0.000000  0.795 0.807
V55LHE   0.003636 0.005729 0.634700 0.354900 0.000000  0.804 0.610
V5GWH    0.003634 0.003729 0.974400 1.325200 1.309900  0.814 0.928
V61SPW   0.003414 0.005979 0.571000 0.000000 0.326200  0.823 0.194
V3GF     0.003253 0.003748 0.868100 1.569300 1.569600  0.831 0.981
V62RBTR  0.003142 0.004683 0.670900 0.127400 0.220000  0.839 0.273
V65LWB   0.002942 0.008165 0.360300 0.258700 0.000000  0.846 0.622
V49LYRE  0.002899 0.005317 0.545200 0.268000 0.000000  0.854 0.707
V66CBW   0.002697 0.004351 0.619700 0.058800 0.220000  0.861 0.388
V85ROSE  0.002535 0.004640 0.546400 0.239700 0.000000  0.867 0.534
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V39FHE   0.002474 0.006936 0.356600 0.248500 0.000000  0.874 0.860
V34WSW   0.002433 0.005328 0.456700 0.247500 0.000000  0.880 0.993
V1GST    0.002432 0.002103 1.156600 1.466600 1.577800  0.886 0.258
V68PIL   0.002421 0.005305 0.456500 0.232900 0.000000  0.892 0.713
V44MUSK  0.002324 0.006421 0.361900 0.236300 0.000000  0.898 0.772
V98FLAME 0.002304 0.005029 0.458100 0.220500 0.000000  0.904 0.783
V87LFC   0.002258 0.004995 0.452000 0.224900 0.000000  0.910 0.653
V51OWH   0.002175 0.004792 0.453900 0.211500 0.000000  0.915 0.464
V52TRM   0.002033 0.005787 0.351300 0.202800 0.000000  0.920 0.750
V47RFC   0.002030 0.004451 0.456200 0.214300 0.000000  0.926 0.906
V30BTR   0.002004 0.005654 0.354500 0.200700 0.000000  0.931 0.939
V25WAG   0.001992 0.004362 0.456700 0.205100 0.000000  0.936 0.994
V38GAL   0.001763 0.004879 0.361300 0.178700 0.000000  0.940 0.973
V77RRP   0.001722 0.004777 0.360400 0.174100 0.000000  0.945 0.345
V76GBB   0.001693 0.003703 0.457300 0.166700 0.000000  0.949 0.639
V86BCOO  0.001443 0.003991 0.361600 0.142300 0.000000  0.953 0.595
V45MGLK  0.001438 0.003967 0.362300 0.142600 0.000000  0.956 0.964
V12LK    0.001432 0.001053 1.360800 1.372500 1.350100  0.960 0.987
V67GGC   0.001406 0.003942 0.356500 0.135400 0.000000  0.964 0.849
V57PINK  0.001334 0.003700 0.360500 0.127300 0.000000  0.967 0.472
V92NFB   0.001291 0.003564 0.362400 0.141400 0.000000  0.970 0.761
V97PCL   0.001110 0.004475 0.248100 0.102200 0.000000  0.973 0.407
V70RSL   0.001028 0.004145 0.248100 0.094600 0.000000  0.976 0.989
V91NMIN  0.000997 0.002784 0.358300 0.103300 0.000000  0.978 0.965
V78LLOR  0.000952 0.003836 0.248100 0.106900 0.000000  0.981 0.423
V60LFB   0.000834 0.003362 0.248100 0.076800 0.000000  0.983 0.956
V84YRTH  0.000776 0.003127 0.248100 0.082700 0.000000  0.985 0.783
V89PCOO  0.000750 0.003025 0.248100 0.069100 0.000000  0.987 0.407
V54STHR  0.000698 0.002812 0.248100 0.073200 0.000000  0.989 0.870
V79YTHE  0.000672 0.002706 0.248100 0.075400 0.000000  0.990 0.759
V82AZKF  0.000607 0.002448 0.248100 0.053800 0.000000  0.992 0.418
V100WBWS 0.000589 0.002375 0.248100 0.066200 0.000000  0.993 0.423
V93DB    0.000578 0.002329 0.248100 0.061600 0.000000  0.995 0.940
V102KING 0.000549 0.002214 0.248100 0.053800 0.000000  0.996 0.606
V90WTG   0.000522 0.002104 0.248100 0.055600 0.000000  0.998 0.974
V71PDOV  0.000495 0.001997 0.248100 0.055600 0.000000  0.999 0.872
V94RBEE  0.000479 0.001931 0.248100 0.053800 0.000000  1.000 0.982
V59YR    0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V72CRP   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V73JW    0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V74BCHE  0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V75RCR   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V88WG    0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V95HBC   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V99WWT   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V101LCOR 0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
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Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Contrast: Mixed_Box-Ironbark

          average       sd    ratio      ava      avb cumsum     p
V18YTH   0.018658 0.005852 3.188000 0.171800 1.909900  0.039 0.001 ***
V26WWCH  0.015902 0.005536 2.872000 0.180300 1.655500  0.072 0.001 ***
V10WBSW  0.015288 0.006045 2.529000 1.426000 0.000000  0.103 0.001 ***
V39FHE   0.014967 0.004566 3.278000 0.248500 1.532700  0.134 0.001 ***
V46BHHE  0.014098 0.007088 1.989000 0.491200 1.798900  0.163 0.002 **
V40BRTH  0.013699 0.008353 1.640000 0.088800 1.379300  0.192 0.005 **
V21ESP   0.013293 0.006583 2.019000 1.236300 0.000000  0.219 0.001 ***
V22SCR   0.012484 0.005255 2.376000 0.210400 1.350800  0.245 0.001 ***
V44MUSK  0.011843 0.007846 1.509000 0.236300 1.180800  0.269 0.005 **
V75RCR   0.010748 0.003075 3.495000 0.000000 1.000900  0.292 0.001 ***
V58OBO   0.009787 0.004514 2.168000 0.198800 1.059200  0.312 0.002 **
V23RBFT  0.009490 0.007770 1.221000 0.990600 0.320500  0.332 0.207
V13RWB   0.008878 0.006402 1.387000 1.080500 1.792200  0.350 0.677
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V56FTC   0.008670 0.006269 1.383000 0.948600 0.281200  0.368 0.040 *
V28VS    0.008644 0.006295 1.373000 0.599300 0.971800  0.386 0.289
V11CR    0.008437 0.007081 1.192000 1.443400 0.992900  0.403 0.769
V36YFHE  0.008272 0.006141 1.347000 1.121400 1.383500  0.420 0.811
V16LR    0.008079 0.006977 1.158000 0.852900 0.576500  0.437 0.543
V35STP   0.007717 0.006654 1.160000 0.700100 0.656700  0.453 0.650
V27NHHE  0.007497 0.008904 0.842000 0.689200 0.000000  0.468 0.670
V19ER    0.007343 0.005945 1.235000 0.816000 0.846400  0.484 0.650
V43GCU   0.007068 0.005570 1.269000 0.423100 0.818400  0.498 0.268
V42SIL   0.006994 0.005548 1.261000 0.952800 1.255000  0.513 0.956
V4BTH    0.006788 0.003994 1.699000 1.946400 1.520900  0.527 0.833
V2EYR    0.006709 0.005999 1.118000 1.285300 0.874900  0.541 0.367
V81SHBC  0.006489 0.005846 1.110000 0.641600 0.281200  0.554 0.398
V66CBW   0.006361 0.006379 0.997000 0.058800 0.566600  0.567 0.008 **
V70RSL   0.006278 0.006199 1.013000 0.094600 0.557400  0.580 0.173
V17WPHE  0.006117 0.009722 0.629000 0.602000 0.000000  0.593 0.981
V80RF    0.006050 0.006578 0.920000 0.568000 0.000000  0.605 0.475
V90WTG   0.006046 0.006464 0.935000 0.055600 0.542000  0.618 0.045 *
V50CHE   0.006025 0.007514 0.802000 0.580200 0.000000  0.630 0.497
V29CST   0.005983 0.006099 0.981000 0.548200 0.256000  0.643 0.928
V61SPW   0.005859 0.005982 0.980000 0.000000 0.593000  0.655 0.025 *
V14AUR   0.005388 0.005537 0.973000 0.893600 1.211600  0.666 0.706
V48YTBC  0.005300 0.006506 0.815000 0.511500 0.000000  0.677 0.536
V15STTH  0.005211 0.006022 0.865000 1.475400 1.627000  0.688 0.971
V9SFW    0.005137 0.005014 1.024000 1.384400 1.472800  0.698 0.596
V41SPP   0.004977 0.006330 0.786000 1.058800 1.440100  0.709 0.829
V30BTR   0.004843 0.007338 0.660000 0.200700 0.382900  0.719 0.544
V5GWH    0.004733 0.003669 1.290000 1.325200 1.171900  0.729 0.707
V3GF     0.004725 0.003539 1.335000 1.569300 1.330200  0.738 0.716
V8WNHE   0.004707 0.005030 0.936000 1.588600 1.595400  0.748 0.915
V37WHIP  0.004647 0.006424 0.723000 0.441900 0.000000  0.758 0.652
V83SFC   0.004543 0.006404 0.710000 0.424600 0.000000  0.767 0.808
V24BFCS  0.004457 0.004309 1.034000 0.751600 1.014400  0.776 0.785
V32SCC   0.004454 0.009873 0.451000 0.441300 0.000000  0.785 0.843
V20PCU   0.004351 0.005983 0.727000 0.411800 0.000000  0.794 0.982
V79YTHE  0.004127 0.006650 0.621000 0.075400 0.396100  0.803 0.027 *
V63DWS   0.004112 0.005037 0.816000 0.313100 0.220000  0.811 0.647
V7WEHE   0.004030 0.003983 1.012000 1.252500 1.296700  0.820 0.962
V33RWH   0.003758 0.004830 0.778000 1.318700 1.512200  0.828 0.996
V6WTTR   0.003705 0.004996 0.741000 1.325800 1.525100  0.835 0.988
V31AMAG  0.003680 0.006826 0.539000 0.370900 0.000000  0.843 0.798
V55LHE   0.003654 0.005776 0.633000 0.354900 0.000000  0.850 0.576
V69SKF   0.003359 0.005292 0.635000 0.334500 0.000000  0.857 0.968
V76GBB   0.003230 0.004486 0.720000 0.166700 0.210200  0.864 0.176
V65LWB   0.002960 0.008240 0.359000 0.258700 0.000000  0.870 0.620
V49LYRE  0.002915 0.005365 0.543000 0.268000 0.000000  0.876 0.713
V96DF    0.002818 0.004531 0.622000 0.070800 0.256000  0.882 0.241
V12LK    0.002704 0.001574 1.718000 1.372500 1.122800  0.888 0.566
V94RBEE  0.002692 0.004346 0.620000 0.053800 0.256000  0.893 0.406
V85ROSE  0.002548 0.004679 0.545000 0.239700 0.000000  0.898 0.512
V34WSW   0.002444 0.005368 0.455000 0.247500 0.000000  0.903 0.995
V68PIL   0.002434 0.005347 0.455000 0.232900 0.000000  0.908 0.692
V53MB    0.002430 0.005496 0.442000 0.224200 0.000000  0.914 0.901
V1GST    0.002363 0.001590 1.487000 1.466600 1.425700  0.918 0.309
V98FLAME 0.002315 0.005070 0.457000 0.220500 0.000000  0.923 0.806
V87LFC   0.002268 0.005033 0.451000 0.224900 0.000000  0.928 0.680
V51OWH   0.002186 0.004829 0.453000 0.211500 0.000000  0.932 0.443
V88WG    0.002185 0.003827 0.571000 0.000000 0.210200  0.937 0.280
V95HBC   0.002185 0.003827 0.571000 0.000000 0.210200  0.942 0.168
V52TRM   0.002043 0.005832 0.350000 0.202800 0.000000  0.946 0.733
V47RFC   0.002039 0.004481 0.455000 0.214300 0.000000  0.950 0.916
V25WAG   0.002001 0.004394 0.455000 0.205100 0.000000  0.954 0.997
V38GAL   0.001771 0.004914 0.360000 0.178700 0.000000  0.958 0.964
V77RRP   0.001730 0.004812 0.359000 0.174100 0.000000  0.961 0.300
V86BCOO  0.001450 0.004021 0.361000 0.142300 0.000000  0.964 0.611
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V45MGLK  0.001444 0.003997 0.361000 0.142600 0.000000  0.967 0.967
V67GGC   0.001413 0.003972 0.356000 0.135400 0.000000  0.970 0.842
V64BELL  0.001400 0.005660 0.247000 0.127500 0.000000  0.973 0.518
V62RBTR  0.001374 0.003843 0.358000 0.127400 0.000000  0.976 0.672
V57PINK  0.001341 0.003729 0.359000 0.127300 0.000000  0.979 0.483
V92NFB   0.001296 0.003585 0.362000 0.141400 0.000000  0.981 0.716
V97PCL   0.001116 0.004512 0.247000 0.102200 0.000000  0.984 0.340
V91NMIN  0.001002 0.002802 0.357000 0.103300 0.000000  0.986 0.965
V78LLOR  0.000955 0.003858 0.248000 0.106900 0.000000  0.988 0.407
V60LFB   0.000839 0.003390 0.247000 0.076800 0.000000  0.990 0.949
V84YRTH  0.000779 0.003146 0.248000 0.082700 0.000000  0.991 0.757
V89PCOO  0.000755 0.003050 0.247000 0.069100 0.000000  0.993 0.340
V54STHR  0.000701 0.002830 0.248000 0.073200 0.000000  0.994 0.839
V82AZKF  0.000611 0.002470 0.247000 0.053800 0.000000  0.995 0.367
V100WBWS 0.000591 0.002388 0.248000 0.066200 0.000000  0.997 0.407
V93DB    0.000580 0.002343 0.248000 0.061600 0.000000  0.998 0.971
V102KING 0.000552 0.002230 0.247000 0.053800 0.000000  0.999 0.570
V71PDOV  0.000497 0.002008 0.248000 0.055600 0.000000  1.000 0.864
V59YR    0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V72CRP   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V73JW    0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V74BCHE  0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V99WWT   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V101LCOR 0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
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Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Contrast: Mixed_River Red Gu

          average       sd    ratio      ava      avb cumsum     p
V4BTH    0.020445 0.006218 3.288000 1.946400 0.000000  0.030 0.001 ***
V32SCC   0.020280 0.007521 2.697000 0.441300 2.225000  0.060 0.002 **
V17WPHE  0.015917 0.009167 1.736000 0.602000 2.034900  0.084 0.018 *
V10WBSW  0.015087 0.005860 2.574000 1.426000 0.000000  0.106 0.001 ***
V11CR    0.015054 0.006597 2.282000 1.443400 0.000000  0.128 0.001 ***
V38GAL   0.014937 0.005417 2.758000 0.178700 1.582800  0.151 0.001 ***
V26WWCH  0.014881 0.004857 3.064000 0.180300 1.578000  0.173 0.003 **
V8WNHE   0.014481 0.007804 1.856000 1.588600 0.289600  0.194 0.003 **
V30BTR   0.014462 0.004730 3.057000 0.200700 1.520300  0.216 0.002 **
V2EYR    0.013450 0.005370 2.505000 1.285300 0.000000  0.236 0.001 ***
V7WEHE   0.013287 0.005547 2.395000 1.252500 0.000000  0.255 0.001 ***
V60LFB   0.013139 0.004117 3.192000 0.076800 1.312300  0.275 0.001 ***
V21ESP   0.013118 0.006422 2.043000 1.236300 0.000000  0.294 0.001 ***
V25WAG   0.012318 0.004959 2.484000 0.205100 1.365200  0.312 0.002 **
V31AMAG  0.011976 0.005536 2.163000 0.370900 1.393900  0.330 0.004 **
V45MGLK  0.011411 0.003990 2.860000 0.142600 1.216900  0.347 0.001 ***
V36YFHE  0.011370 0.007879 1.443000 1.121400 0.000000  0.364 0.034 *
V13RWB   0.011326 0.007442 1.522000 1.080500 0.000000  0.381 0.054 .
V47RFC   0.011238 0.004799 2.342000 0.214300 1.251400  0.397 0.002 **
V70RSL   0.011018 0.007241 1.522000 0.094600 1.085200  0.414 0.004 **
V41SPP   0.010892 0.006237 1.746000 1.058800 0.000000  0.430 0.006 **
V15STTH  0.010499 0.008202 1.280000 1.475400 0.745400  0.445 0.139
V35STP   0.010090 0.007467 1.351000 0.700100 1.638300  0.460 0.032 *
V73JW    0.009687 0.005914 1.638000 0.000000 0.934100  0.474 0.001 ***
V33RWH   0.009674 0.006163 1.570000 1.318700 0.532700  0.489 0.032 *
V42SIL   0.009667 0.007712 1.253000 0.952800 0.442300  0.503 0.118
V34WSW   0.009358 0.006570 1.424000 0.247500 0.961700  0.517 0.130
V52TRM   0.009124 0.005824 1.567000 0.202800 0.871300  0.531 0.005 **
V98FLAME 0.009079 0.006396 1.419000 0.220500 0.964700  0.544 0.006 **
V69SKF   0.008992 0.005367 1.675000 0.334500 1.156400  0.557 0.010 **
V18YTH   0.008930 0.008900 1.003000 0.171800 0.845000  0.571 0.294
V23RBFT  0.008751 0.006449 1.357000 0.990600 0.456500  0.584 0.393
V6WTTR   0.008747 0.007272 1.203000 1.325800 0.736100  0.597 0.180
V56FTC   0.008722 0.005609 1.555000 0.948600 0.220000  0.609 0.035 *
V93DB    0.008385 0.005276 1.589000 0.061600 0.829800  0.622 0.002 **
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V91NMIN  0.008172 0.007902 1.034000 0.103300 0.767400  0.634 0.026 *
V28VS    0.007982 0.007161 1.115000 0.599300 0.686000  0.646 0.629
V5GWH    0.007767 0.005012 1.550000 1.325200 0.765000  0.657 0.088 .
V94RBEE  0.007444 0.004665 1.596000 0.053800 0.732200  0.668 0.004 **
V27NHHE  0.007397 0.008745 0.846000 0.689200 0.000000  0.679 0.669
V19ER    0.007375 0.006436 1.146000 0.816000 1.397300  0.690 0.663
V16LR    0.006856 0.005658 1.212000 0.852900 1.252000  0.700 0.857
V59YR    0.006854 0.006952 0.986000 0.000000 0.651800  0.711 0.002 **
V81SHBC  0.006705 0.005721 1.172000 0.641600 0.000000  0.720 0.258
V3GF     0.006585 0.003090 2.131000 1.569300 1.077300  0.730 0.285
V29CST   0.006553 0.005132 1.277000 0.548200 0.776000  0.740 0.645
V20PCU   0.006505 0.006055 1.074000 0.411800 0.602800  0.750 0.532
V88WG    0.006448 0.006586 0.979000 0.000000 0.612900  0.759 0.003 **
V90WTG   0.006446 0.006479 0.995000 0.055600 0.612900  0.769 0.034 *
V99WWT   0.006439 0.006551 0.983000 0.000000 0.615300  0.778 0.008 **
V58OBO   0.006407 0.006318 1.014000 0.198800 0.611700  0.788 0.100 .
V24BFCS  0.006242 0.005422 1.151000 0.751600 1.328700  0.797 0.201
V9SFW    0.006159 0.003969 1.552000 1.384400 1.145500  0.806 0.352
V80RF    0.005971 0.006465 0.924000 0.568000 0.000000  0.815 0.505
V50CHE   0.005950 0.007394 0.805000 0.580200 0.000000  0.824 0.502
V92NFB   0.005676 0.005583 1.017000 0.141400 0.513100  0.832 0.023 *
V72CRP   0.005564 0.005644 0.986000 0.000000 0.545600  0.840 0.009 **
V71PDOV  0.005528 0.005717 0.967000 0.055600 0.530500  0.849 0.015 *
V48YTBC  0.005233 0.006402 0.817000 0.511500 0.000000  0.856 0.571
V14AUR   0.005141 0.005536 0.929000 0.893600 1.237700  0.864 0.755
V12LK    0.005031 0.005700 0.883000 1.372500 0.910300  0.871 0.130
V46BHHE  0.004971 0.006225 0.799000 0.491200 0.000000  0.879 0.998
V37WHIP  0.004588 0.006321 0.726000 0.441900 0.000000  0.885 0.675
V84YRTH  0.004512 0.007311 0.617000 0.082700 0.389600  0.892 0.027 *
V83SFC   0.004484 0.006299 0.712000 0.424600 0.000000  0.899 0.820
V43GCU   0.004425 0.006145 0.720000 0.423100 0.000000  0.905 0.980
V87LFC   0.004386 0.006114 0.717000 0.224900 0.314400  0.912 0.212
V53MB    0.003834 0.005402 0.710000 0.224200 0.210200  0.917 0.646
V101LCOR 0.003699 0.006477 0.571000 0.000000 0.370000  0.923 0.072 .
V55LHE   0.003608 0.005688 0.634000 0.354900 0.000000  0.928 0.593
V63DWS   0.003160 0.005040 0.627000 0.313100 0.000000  0.933 0.851
V65LWB   0.002918 0.008100 0.360000 0.258700 0.000000  0.937 0.596
V49LYRE  0.002876 0.005277 0.545000 0.268000 0.000000  0.942 0.712
V74BCHE  0.002855 0.005003 0.571000 0.000000 0.261700  0.946 0.065 .
V85ROSE  0.002515 0.004606 0.546000 0.239700 0.000000  0.950 0.550
V39FHE   0.002455 0.006886 0.357000 0.248500 0.000000  0.953 0.897
V68PIL   0.002403 0.005266 0.456000 0.232900 0.000000  0.957 0.709
V44MUSK  0.002307 0.006375 0.362000 0.236300 0.000000  0.960 0.800
V95HBC   0.002170 0.003801 0.571000 0.000000 0.198800  0.963 0.175
V51OWH   0.002158 0.004757 0.454000 0.211500 0.000000  0.967 0.454
V22SCR   0.002084 0.004592 0.454000 0.210400 0.000000  0.970 1.000
V1GST    0.002006 0.001368 1.466000 1.466600 1.343300  0.973 0.616
V77RRP   0.001709 0.004743 0.360000 0.174100 0.000000  0.975 0.380
V76GBB   0.001680 0.003676 0.457000 0.166700 0.000000  0.978 0.677
V86BCOO  0.001432 0.003963 0.361000 0.142300 0.000000  0.980 0.620
V67GGC   0.001395 0.003914 0.356000 0.135400 0.000000  0.982 0.860
V64BELL  0.001382 0.005570 0.248000 0.127500 0.000000  0.984 0.559
V62RBTR  0.001356 0.003781 0.359000 0.127400 0.000000  0.986 0.709
V57PINK  0.001324 0.003673 0.360000 0.127300 0.000000  0.988 0.485
V97PCL   0.001101 0.004441 0.248000 0.102200 0.000000  0.990 0.443
V40BRTH  0.001017 0.004100 0.248000 0.088800 0.000000  0.991 0.999
V78LLOR  0.000946 0.003812 0.248000 0.106900 0.000000  0.992 0.474
V89PCOO  0.000744 0.003002 0.248000 0.069100 0.000000  0.994 0.443
V54STHR  0.000693 0.002793 0.248000 0.073200 0.000000  0.995 0.859
V79YTHE  0.000667 0.002689 0.248000 0.075400 0.000000  0.996 0.803
V66CBW   0.000652 0.002628 0.248000 0.058800 0.000000  0.997 0.916
V96DF    0.000626 0.002525 0.248000 0.070800 0.000000  0.997 0.847
V82AZKF  0.000602 0.002428 0.248000 0.053800 0.000000  0.998 0.475
V100WBWS 0.000585 0.002359 0.248000 0.066200 0.000000  0.999 0.474
V102KING 0.000545 0.002198 0.248000 0.053800 0.000000  1.000 0.603
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V61SPW   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V75RCR   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
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Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Contrast: Gippsland Ma_Montane Fore

           average       sd     ratio       ava       avb cumsum     p
V17WPHE   0.025511  0.002976  8.572000  2.247700  0.000000  0.047 0.002 **
V34WSW    0.020923  0.002668  7.841000  1.856000  0.000000  0.086 0.001 ***
V11CR     0.019365  0.002542  7.617000  0.000000  1.710400  0.121 0.002 **
V27NHHE   0.018629  0.005197  3.584000  1.646300  0.000000  0.156 0.002 **
V6WTTR    0.017764  0.002227  7.977000  0.000000  1.567100  0.189 0.002 **
V15STTH   0.017530  0.007233  2.423000  0.323400  1.857500  0.221 0.005 **
V25WAG    0.015594  0.002143  7.276000  1.376000  0.000000  0.250 0.001 ***
V13RWB    0.014804  0.009495  1.559000  1.922100  0.662400  0.277 0.011 *
V20PCU    0.014406  0.001861  7.739000  0.000000  1.268100  0.304 0.003 **
V43GCU    0.014108  0.001624  8.689000  0.000000  1.256100  0.330 0.002 **
V83SFC    0.013763  0.001275 10.794000  0.000000  1.220000  0.355 0.002 **
V4BTH     0.012989  0.009916  1.310000  1.184200  2.294000  0.379 0.102
V42SIL    0.012586  0.005694  2.210000  0.420400  1.363600  0.402 0.034 *
V3GF      0.012521  0.008426  1.486000  0.646500  1.724600  0.425 0.007 **
V19ER     0.011718  0.005658  2.071000  1.308600  0.266900  0.447 0.029 *
V37WHIP   0.011155  0.006729  1.658000  0.000000  1.020400  0.468 0.003 **
V48YTBC   0.010991  0.006611  1.663000  0.000000  0.958400  0.488 0.006 **
V81SHBC   0.010480  0.005715  1.834000  0.271900  1.191300  0.507 0.012 *
V21ESP    0.010274  0.006547  1.569000  0.644000  1.502100  0.526 0.142
V67GGC    0.010101  0.006264  1.613000  0.000000  0.926600  0.545 0.002 **
V80RF     0.010024  0.006191  1.619000  0.000000  0.913500  0.563 0.026 *
V5GWH     0.009874  0.007909  1.248000  0.674800  1.486100  0.582 0.028 *
V29CST    0.009751  0.004639  2.102000  1.059100  0.314400  0.599 0.047 *
V36YFHE   0.009681  0.007030  1.377000  1.243700  1.134500  0.617 0.410
V16LR     0.009375  0.007159  1.310000  1.184900  1.023200  0.635 0.243
V46BHHE   0.008981  0.008370  1.073000  0.307900  0.828800  0.651 0.389
V49LYRE   0.008738  0.005375  1.626000  0.000000  0.803100  0.667 0.013 *
V33RWH    0.008651  0.007180  1.205000  0.579800  0.753700  0.683 0.264
V68PIL    0.008604  0.005317  1.618000  0.000000  0.785200  0.699 0.012 *
V54STHR   0.008499  0.005150  1.650000  0.000000  0.758900  0.715 0.001 ***
V65LWB    0.008071  0.008400  0.961000  0.741500  0.000000  0.730 0.153
V63DWS    0.007899  0.008217  0.961000  0.724800  0.000000  0.744 0.063 .
V91NMIN   0.007851  0.008209  0.956000  0.719700  0.000000  0.759 0.081 .
V28VS     0.007829  0.008154  0.960000  0.000000  0.709200  0.773 0.611
V7WEHE    0.007421  0.005457  1.360000  1.241500  1.224500  0.787 0.378
V24BFCS   0.007277  0.005961  1.221000  1.112500  0.496700  0.800 0.129
V56FTC    0.007098  0.006534  1.086000  0.615700  1.231300  0.813 0.428
V35STP    0.006718  0.007148  0.940000  0.993200  0.944200  0.826 0.805
V23RBFT   0.006640  0.006306  1.053000  0.937700  0.883700  0.838 0.889
V14AUR    0.006610  0.006785  0.974000  0.962100  0.919800  0.850 0.421
V55LHE    0.006240  0.006468  0.965000  0.000000  0.590500  0.862 0.186
V12LK     0.005918  0.005486  1.079000  0.801000  1.324100  0.873 0.071 .
V26WWCH   0.005801  0.010413  0.557000  0.000000  0.462400  0.883 0.892
V50CHE    0.005678  0.005941  0.956000  0.000000  0.509600  0.894 0.587
V86BCOO   0.005379  0.005588  0.962000  0.000000  0.481700  0.904 0.081 .
V22SCR    0.003983  0.007149  0.557000  0.000000  0.317500  0.911 0.851
V8WNHE    0.003758  0.001583  2.374000  1.843200  1.709100  0.918 0.877
V85ROSE   0.003564  0.006391  0.558000  0.000000  0.331700  0.925 0.300
V62RBTR   0.003528  0.006332  0.557000  0.000000  0.281200  0.931 0.212
V38GAL    0.003497  0.006277  0.557000  0.323400  0.000000  0.938 0.801
V87LFC    0.003183  0.005712  0.557000  0.289600  0.000000  0.944 0.573
V45MGLK   0.002941  0.005278  0.557000  0.271900  0.000000  0.949 0.813
V41SPP    0.002919  0.001917  1.523000  1.200500  1.352200  0.954 0.890
V9SFW     0.002819  0.002623  1.075000  1.734700  1.525000  0.960 0.917
V53MB     0.002704  0.004852  0.557000  0.250000  0.000000  0.965 0.808
V57PINK   0.002686  0.004816  0.558000  0.000000  0.250000  0.970 0.270
V76GBB    0.002568  0.004606  0.557000  0.000000  0.220000  0.974 0.381
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V70RSL    0.002557  0.004589  0.557000  0.236400  0.000000  0.979 0.799
V2EYR     0.002508  0.001941  1.292000  1.599500  1.440100  0.984 0.914
V102KING  0.002459  0.004408  0.558000  0.000000  0.236400  0.988 0.069 .
V51OWH    0.002364  0.004239  0.558000  0.000000  0.220000  0.993 0.535
V10WBSW   0.002324  0.001192  1.950000  1.760400  1.622600  0.997 0.968
V1GST     0.001754  0.001373  1.278000  1.493300  1.379200  1.000 0.741
V18YTH    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V30BTR    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V31AMAG   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V32SCC    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V39FHE    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V40BRTH   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V44MUSK   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V47RFC    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V52TRM    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V58OBO    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V59YR     0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V60LFB    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V61SPW    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V64BELL   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V66CBW    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V69SKF    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V71PDOV   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V72CRP    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V73JW     0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V74BCHE   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V75RCR    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V77RRP    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V78LLOR   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V79YTHE   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V82AZKF   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V84YRTH   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V88WG     0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V89PCOO   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V90WTG    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V92NFB    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V93DB     0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V94RBEE   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V95HBC    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V96DF     0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V97PCL    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V98FLAME  0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V99WWT    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V100WBWS  0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V101LCOR  0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
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Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Contrast: Gippsland Ma_Foothills Wo

           average        sd     ratio       ava       avb cumsum     p
V17WPHE   0.026097  0.002383 10.951000  2.247700  0.000000  0.052 0.001 ***
V34WSW    0.021402  0.002230  9.597000  1.856000  0.000000  0.094 0.001 ***
V6WTTR    0.018324  0.001427 12.839000  0.000000  1.579900  0.130 0.001 ***
V11CR     0.017226  0.001813  9.499000  0.000000  1.484400  0.164 0.003 **
V15STTH   0.017112  0.007416  2.307000  0.323400  1.788200  0.198 0.003 **
V40BRTH   0.016327  0.010384  1.572000  0.000000  1.423800  0.230 0.003 **
V27NHHE   0.015988  0.007215  2.216000  1.646300  0.256000  0.262 0.012 *
V25WAG    0.015952  0.001857  8.589000  1.376000  0.000000  0.293 0.001 ***
V22SCR    0.015801  0.001670  9.460000  0.000000  1.364500  0.325 0.001 ***
V26WWCH   0.013884  0.008667  1.602000  0.000000  1.210300  0.352 0.018 *
V18YTH    0.013759  0.008550  1.609000  0.000000  1.199300  0.379 0.021 *
V28VS     0.012735  0.007685  1.657000  0.000000  1.088900  0.404 0.016 *
V10WBSW   0.011739  0.009043  1.298000  1.760400  0.773600  0.428 0.062 .
V3GF      0.011589  0.007893  1.468000  0.646500  1.569600  0.451 0.010 **
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V33RWH    0.011462  0.006709  1.709000  0.579800  1.597500  0.473 0.016 *
V69SKF    0.010219  0.006162  1.658000  0.000000  0.890600  0.494 0.008 **
V4BTH     0.010179  0.009715  1.048000  1.184200  1.975000  0.514 0.349
V23RBFT   0.009465  0.006668  1.419000  0.937700  0.243500  0.532 0.315
V46BHHE   0.009439  0.007230  1.305000  0.307900  0.931700  0.551 0.278
V42SIL    0.009295  0.005353  1.736000  0.420400  0.761200  0.569 0.289
V21ESP    0.009082  0.006693  1.357000  0.644000  1.346100  0.587 0.294
V5GWH     0.008800  0.007323  1.202000  0.674800  1.309900  0.605 0.082 .
V13RWB    0.008699  0.002320  3.750000  1.922100  1.170600  0.622 0.646
V53MB     0.008525  0.006365  1.339000  0.250000  0.851600  0.639 0.026 *
V36YFHE   0.008313  0.004873  1.706000  1.243700  1.319700  0.655 0.696
V65LWB    0.008250  0.008548  0.965000  0.741500  0.000000  0.671 0.149
V63DWS    0.008179  0.007487  1.092000  0.724800  0.243500  0.688 0.041 *
V81SHBC   0.008171  0.006178  1.323000  0.271900  0.807100  0.704 0.076 .
V19ER     0.008140  0.007125  1.143000  1.308600  0.654600  0.720 0.354
V91NMIN   0.008026  0.008354  0.961000  0.719700  0.000000  0.736 0.062 .
V7WEHE    0.007740  0.006507  1.189000  1.241500  1.318200  0.751 0.319
V29CST    0.007736  0.004933  1.568000  1.059100  0.659900  0.766 0.202
V20PCU    0.007716  0.008005  0.964000  0.000000  0.673000  0.782 0.170
V9SFW     0.007643  0.007705  0.992000  1.734700  1.081900  0.797 0.205
V56FTC    0.007448  0.006647  1.120000  0.615700  1.244700  0.811 0.262
V14AUR    0.006977  0.007154  0.975000  0.962100  0.987000  0.825 0.356
V35STP    0.006914  0.007551  0.916000  0.993200  1.011200  0.839 0.760
V43GCU    0.006636  0.007051  0.941000  0.000000  0.583000  0.852 0.494
V12LK     0.006350  0.005685  1.117000  0.801000  1.350100  0.865 0.044 *
V16LR     0.006246  0.006500  0.961000  1.184900  1.479800  0.877 0.839
V64BELL   0.005568  0.009988  0.557000  0.000000  0.492000  0.888 0.215
V50CHE    0.004608  0.008269  0.557000  0.000000  0.384600  0.897 0.689
V96DF     0.004006  0.007186  0.557000  0.000000  0.349000  0.905 0.103
V41SPP    0.003877  0.002906  1.334000  1.200500  1.526000  0.913 0.886
V24BFCS   0.003777  0.005287  0.714000  1.112500  0.794400  0.920 0.836
V61SPW    0.003744  0.006716  0.557000  0.000000  0.326200  0.927 0.117
V80RF     0.003648  0.006547  0.557000  0.000000  0.304500  0.935 0.850
V2EYR     0.003593  0.002576  1.394000  1.599500  1.316800  0.942 0.853
V38GAL    0.003574  0.006396  0.559000  0.323400  0.000000  0.949 0.765
V87LFC    0.003254  0.005822  0.559000  0.289600  0.000000  0.955 0.541
V8WNHE    0.003139  0.002678  1.172000  1.843200  1.582200  0.962 0.917
V58OBO    0.003041  0.005456  0.557000  0.000000  0.261700  0.968 0.800
V83SFC    0.003041  0.005456  0.557000  0.000000  0.261700  0.974 0.867
V45MGLK   0.003006  0.005378  0.559000  0.271900  0.000000  0.980 0.756
V1GST     0.002663  0.001997  1.334000  1.493300  1.577800  0.985 0.199
V70RSL    0.002613  0.004676  0.559000  0.236400  0.000000  0.990 0.807
V62RBTR   0.002557  0.004588  0.557000  0.000000  0.220000  0.995 0.452
V66CBW    0.002557  0.004588  0.557000  0.000000  0.220000  1.000 0.528
V30BTR    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V31AMAG   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V32SCC    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V37WHIP   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V39FHE    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V44MUSK   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V47RFC    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V48YTBC   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V49LYRE   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V51OWH    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V52TRM    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V54STHR   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V55LHE    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V57PINK   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V59YR     0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V60LFB    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V67GGC    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V68PIL    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V71PDOV   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V72CRP    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V73JW     0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
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V74BCHE   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V75RCR    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V76GBB    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V77RRP    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V78LLOR   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V79YTHE   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V82AZKF   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V84YRTH   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V85ROSE   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V86BCOO   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V88WG     0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V89PCOO   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V90WTG    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V92NFB    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V93DB     0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V94RBEE   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V95HBC    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V97PCL    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V98FLAME  0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V99WWT    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V100WBWS  0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V101LCOR  0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V102KING  0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
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Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Contrast: Gippsland Ma_Box-Ironbark

          average       sd    ratio      ava      avb cumsum     p
V17WPHE  0.026261 0.003224 8.146000 2.247700 0.000000  0.046 0.001 ***
V18YTH   0.022239 0.002654 8.378000 0.000000 1.909900  0.086 0.001 ***
V34WSW   0.021534 0.002823 7.627000 1.856000 0.000000  0.124 0.001 ***
V10WBSW  0.020600 0.003335 6.178000 1.760400 0.000000  0.160 0.001 ***
V26WWCH  0.019377 0.002919 6.638000 0.000000 1.655500  0.195 0.001 ***
V27NHHE  0.019175 0.005389 3.559000 1.646300 0.000000  0.229 0.002 **
V6WTTR   0.017796 0.001927 9.234000 0.000000 1.525100  0.260 0.002 **
V46BHHE  0.017556 0.007155 2.454000 0.307900 1.798900  0.291 0.001 ***
V39FHE   0.017550 0.003257 5.388000 0.000000 1.532700  0.322 0.001 ***
V25WAG   0.016052 0.002282 7.033000 1.376000 0.000000  0.351 0.001 ***
V22SCR   0.015788 0.003354 4.707000 0.000000 1.350800  0.379 0.001 ***
V40BRTH  0.015511 0.009337 1.661000 0.000000 1.379300  0.406 0.004 **
V15STTH  0.015362 0.007523 2.042000 0.323400 1.627000  0.433 0.013 *
V44MUSK  0.013318 0.008639 1.542000 0.000000 1.180800  0.457 0.005 **
V58OBO   0.012439 0.003104 4.007000 0.000000 1.059200  0.479 0.001 ***
V42SIL   0.012248 0.004880 2.510000 0.420400 1.255000  0.500 0.028 *
V75RCR   0.011824 0.003514 3.365000 0.000000 1.000900  0.521 0.001 ***
V28VS    0.011169 0.006772 1.649000 0.000000 0.971800  0.541 0.064 .
V11CR    0.011098 0.006986 1.589000 0.000000 0.992900  0.561 0.153
V33RWH   0.010578 0.006786 1.559000 0.579800 1.512200  0.580 0.038 *
V16LR    0.010525 0.007973 1.320000 1.184900 0.576500  0.598 0.102
V3GF     0.010144 0.006630 1.530000 0.646500 1.330200  0.616 0.032 *
V29CST   0.009753 0.005485 1.778000 1.059100 0.256000  0.633 0.039 *
V23RBFT  0.009549 0.007039 1.357000 0.937700 0.320500  0.650 0.318
V43GCU   0.009226 0.005611 1.644000 0.000000 0.818400  0.667 0.058 .
V35STP   0.009032 0.007151 1.263000 0.993200 0.656700  0.683 0.229
V36YFHE  0.008694 0.006441 1.350000 1.243700 1.383500  0.698 0.625
V2EYR    0.008634 0.006941 1.244000 1.599500 0.874900  0.713 0.153
V5GWH    0.008626 0.006596 1.308000 0.674800 1.171900  0.729 0.094 .
V65LWB   0.008297 0.008644 0.960000 0.741500 0.000000  0.743 0.145
V63DWS   0.008199 0.007646 1.072000 0.724800 0.220000  0.758 0.048 *
V91NMIN  0.008072 0.008448 0.955000 0.719700 0.000000  0.772 0.063 .
V21ESP   0.007818 0.008323 0.939000 0.644000 0.000000  0.786 0.681
V7WEHE   0.007692 0.006430 1.196000 1.241500 1.296700  0.799 0.331
V4BTH    0.007495 0.007712 0.972000 1.184200 1.520900  0.813 0.753
V56FTC   0.007373 0.007510 0.982000 0.615700 0.281200  0.826 0.303
```

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V66CBW   0.007079 0.007358 0.962000 0.000000 0.566600  0.838 0.003 **
V90WTG   0.006660 0.007437 0.896000 0.000000 0.542000  0.850 0.037 *
V70RSL   0.006539 0.006415 1.019000 0.236400 0.557400  0.862 0.242
V61SPW   0.006393 0.006678 0.957000 0.000000 0.593000  0.873 0.027 *
V19ER    0.006130 0.006577 0.932000 1.308600 0.846400  0.884 0.860
V14AUR   0.005432 0.006537 0.831000 0.962100 1.211600  0.893 0.627
V81SHBC  0.004610 0.006122 0.753000 0.271900 0.281200  0.902 0.940
V12LK    0.004495 0.005039 0.892000 0.801000 1.122800  0.910 0.303
V79YTHE  0.004087 0.007327 0.558000 0.000000 0.396100  0.917 0.052 .
V9SFW    0.004012 0.003799 1.056000 1.734700 1.472800  0.924 0.745
V30BTR   0.003950 0.007083 0.558000 0.000000 0.382900  0.931 0.653
V38GAL   0.003595 0.006455 0.557000 0.323400 0.000000  0.937 0.754
V13RWB   0.003563 0.001691 2.107000 1.922100 1.792200  0.944 0.999
V41SPP   0.003386 0.002315 1.463000 1.200500 1.440100  0.950 0.886
V87LFC   0.003273 0.005878 0.557000 0.289600 0.000000  0.955 0.500
V45MGLK  0.003023 0.005428 0.557000 0.271900 0.000000  0.961 0.764
V8WNHE   0.002915 0.000912 3.194000 1.843200 1.595400  0.966 0.933
V53MB    0.002779 0.004990 0.557000 0.250000 0.000000  0.971 0.763
V94RBEE  0.002642 0.004736 0.558000 0.000000 0.256000  0.976 0.426
V96DF    0.002642 0.004736 0.558000 0.000000 0.256000  0.980 0.400
V76GBB   0.002548 0.004573 0.557000 0.000000 0.210200  0.985 0.422
V1GST    0.002444 0.001755 1.393000 1.493300 1.425700  0.989 0.334
V88WG    0.002395 0.004296 0.557000 0.000000 0.210200  0.993 0.392
V95HBC   0.002395 0.004296 0.557000 0.000000 0.210200  0.997 0.051 .
V24BFCS  0.001441 0.001298 1.110000 1.112500 1.014400  1.000 0.985
V20PCU   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V31AMAG  0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V32SCC   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V37WHIP  0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V47RFC   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V48YTBC  0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V49LYRE  0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V50CHE   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V51OWH   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V52TRM   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V54STHR  0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V55LHE   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V57PINK  0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V59YR    0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V60LFB   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V62RBTR  0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V64BELL  0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V67GGC   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V68PIL   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V69SKF   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V71PDOV  0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V72CRP   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V73JW    0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V74BCHE  0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V77RRP   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V78LLOR  0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V80RF    0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V82AZKF  0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V83SFC   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V84YRTH  0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V85ROSE  0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V86BCOO  0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V89PCOO  0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V92NFB   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V93DB    0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V97PCL   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V98FLAME 0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V99WWT   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V100WBWS 0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V101LCOR 0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
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V102KING 0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
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Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Contrast: Gippsland Ma_River Red Gu

           average        sd     ratio       ava       avb cumsum     p
V32SCC    0.025596  0.002660  9.624000  0.000000  2.225000  0.041 0.001 ***
V13RWB    0.022078  0.001581 13.960000  1.922100  0.000000  0.077 0.001 ***
V10WBSW   0.020296  0.002871  7.069000  1.760400  0.000000  0.110 0.001 ***
V27NHHE   0.018894  0.005104  3.702000  1.646300  0.000000  0.140 0.002 **
V2EYR     0.018457  0.002843  6.492000  1.599500  0.000000  0.170 0.001 ***
V26WWCH   0.018125  0.001284 14.119000  0.000000  1.578000  0.199 0.004 **
V8WNHE    0.018039  0.006464  2.791000  1.843200  0.289600  0.228 0.001 ***
V30BTR    0.017484  0.001776  9.844000  0.000000  1.520300  0.257 0.001 ***
V31AMAG   0.016036  0.001581 10.146000  0.000000  1.393900  0.283 0.002 **
V60LFB    0.015147  0.003696  4.098000  0.000000  1.312300  0.307 0.001 ***
V38GAL    0.014698  0.007189  2.044000  0.323400  1.582800  0.331 0.002 **
V47RFC    0.014352  0.001420 10.106000  0.000000  1.251400  0.354 0.001 ***
V36YFHE   0.014260  0.009360  1.524000  1.243700  0.000000  0.377 0.018 *
V7WEHE    0.013756  0.008586  1.602000  1.241500  0.000000  0.399 0.004 **
V41SPP    0.013744  0.002390  5.750000  1.200500  0.000000  0.421 0.003 **
V69SKF    0.013273  0.001217 10.902000  0.000000  1.156400  0.443 0.001 ***
V4BTH     0.013108  0.008293  1.581000  1.184200  0.000000  0.464 0.071 .
V70RSL    0.011108  0.007818  1.421000  0.236400  1.085200  0.482 0.010 **
V45MGLK   0.010974  0.005854  1.875000  0.271900  1.216900  0.500 0.002 **
V98FLAME  0.010959  0.006960  1.575000  0.000000  0.964700  0.517 0.001 ***
V73JW     0.010616  0.006635  1.600000  0.000000  0.934100  0.534 0.001 ***
V34WSW    0.010143  0.007370  1.376000  1.856000  0.961700  0.551 0.158
V52TRM    0.009996  0.006422  1.556000  0.000000  0.871300  0.567 0.012 *
V18YTH    0.009754  0.010215  0.955000  0.000000  0.845000  0.583 0.267
V93DB     0.009507  0.005742  1.656000  0.000000  0.829800  0.598 0.001 ***
V91NMIN   0.008898  0.008710  1.022000  0.719700  0.767400  0.612 0.024 *
V15STTH   0.008597  0.008559  1.004000  0.323400  0.745400  0.626 0.477
V3GF      0.008595  0.004954  1.735000  0.646500  1.077300  0.640 0.097 .
V6WTTR    0.008498  0.008891  0.956000  0.000000  0.736100  0.654 0.328
V94RBEE   0.008405  0.005103  1.647000  0.000000  0.732200  0.667 0.001 ***
V65LWB    0.008183  0.008484  0.965000  0.741500  0.000000  0.681 0.150
V23RBFT   0.008089  0.005959  1.357000  0.937700  0.456500  0.694 0.600
V63DWS    0.008008  0.008299  0.965000  0.724800  0.000000  0.707 0.051 .
V28VS     0.007914  0.008215  0.963000  0.000000  0.686000  0.720 0.576
V35STP    0.007822  0.007933  0.986000  0.993200  1.638300  0.732 0.582
V5GWH     0.007810  0.005625  1.388000  0.674800  0.765000  0.745 0.168
V21ESP    0.007696  0.008143  0.945000  0.644000  0.000000  0.757 0.715
V59YR     0.007520  0.007810  0.963000  0.000000  0.651800  0.769 0.001 ***
V42SIL    0.007277  0.009602  0.758000  0.420400  0.442300  0.781 0.755
V56FTC    0.007269  0.006882  1.056000  0.615700  0.220000  0.793 0.338
V88WG     0.007075  0.007399  0.956000  0.000000  0.612900  0.804 0.002 **
V90WTG    0.007075  0.007399  0.956000  0.000000  0.612900  0.816 0.028 *
V99WWT    0.007063  0.007361  0.959000  0.000000  0.615300  0.827 0.001 ***
V58OBO    0.007005  0.007268  0.964000  0.000000  0.611700  0.838 0.115
V9SFW     0.006866  0.003914  1.754000  1.734700  1.145500  0.849 0.316
V33RWH    0.006745  0.006466  1.043000  0.579800  0.532700  0.860 0.625
V20PCU    0.006711  0.007071  0.949000  0.000000  0.602800  0.871 0.510
V16LR     0.006244  0.004942  1.264000  1.184900  1.252000  0.881 0.858
V72CRP    0.006087  0.006321  0.963000  0.000000  0.545600  0.891 0.001 ***
V92NFB    0.006062  0.006313  0.960000  0.000000  0.513100  0.901 0.007 **
V71PDOV   0.006011  0.006489  0.926000  0.000000  0.530500  0.910 0.001 ***
V12LK     0.005937  0.005887  1.009000  0.801000  0.910300  0.920 0.057 .
V14AUR    0.005129  0.006478  0.792000  0.962100  1.237700  0.928 0.665
V87LFC    0.005124  0.006716  0.763000  0.289600  0.314400  0.937 0.206
V84YRTH   0.004535  0.008137  0.557000  0.000000  0.389600  0.944 0.045 *
V101LCOR  0.004040  0.007245  0.558000  0.000000  0.370000  0.951 0.001 ***
V53MB     0.004022  0.005167  0.778000  0.250000  0.210200  0.957 0.594
V29CST    0.003883  0.005041  0.770000  1.059100  0.776000  0.963 0.975
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V46BHHE   0.003418  0.006118  0.559000  0.307900  0.000000  0.969 0.990
V74BCHE   0.003144  0.005642  0.557000  0.000000  0.261700  0.974 0.002 **
V81SHBC   0.003030  0.005425  0.559000  0.271900  0.000000  0.979 0.987
V17WPHE   0.002749  0.001706  1.611000  2.247700  2.034900  0.983 0.982
V24BFCS   0.002646  0.002193  1.206000  1.112500  1.328700  0.988 0.904
V95HBC    0.002389  0.004287  0.557000  0.000000  0.198800  0.991 0.047 *
V1GST     0.002019  0.001634  1.236000  1.493300  1.343300  0.995 0.553
V19ER     0.001923  0.001172  1.640000  1.308600  1.397300  0.998 0.996
V25WAG    0.001422  0.001101  1.292000  1.376000  1.365200  1.000 0.959
V11CR     0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V22SCR    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V37WHIP   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V39FHE    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V40BRTH   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V43GCU    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V44MUSK   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V48YTBC   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V49LYRE   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V50CHE    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V51OWH    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V54STHR   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V55LHE    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V57PINK   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V61SPW    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V62RBTR   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V64BELL   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V66CBW    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V67GGC    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V68PIL    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V75RCR    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V76GBB    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V77RRP    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V78LLOR   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V79YTHE   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V80RF     0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V82AZKF   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V83SFC    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V85ROSE   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V86BCOO   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V89PCOO   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V96DF     0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V97PCL    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V100WBWS  0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V102KING  0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
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Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Contrast: Montane Fore_Foothills Wo 

          average       sd    ratio      ava      avb cumsum     p    
V40BRTH  0.014435 0.009183 1.571800 0.000000 1.423800  0.042 0.015 *
V18YTH   0.012164 0.007560 1.609000 0.000000 1.199300  0.077 0.075 .
V26WWCH  0.010745 0.008003 1.342600 0.462400 1.210300  0.108 0.142
V22SCR   0.010509 0.005505 1.909200 0.317500 1.364500  0.138 0.030 *
V37WHIP  0.010114 0.006054 1.670700 1.020400 0.000000  0.168 0.021 *
V48YTBC  0.009915 0.005918 1.675400 0.958400 0.000000  0.196 0.027 *
V83SFC   0.009752 0.004877 1.999500 1.220000 0.261700  0.224 0.029 *
V10WBSW  0.009395 0.007408 1.268200 1.622600 0.773600  0.251 0.411
V67GGC   0.009162 0.005646 1.622600 0.926600 0.000000  0.278 0.006 **
V69SKF   0.009034 0.005447 1.658400 0.000000 0.890600  0.304 0.039 *
V53MB    0.008754 0.005324 1.644300 0.000000 0.851600  0.329 0.017 *
V33RWH   0.008585 0.007066 1.215100 0.753700 1.597500  0.354 0.271
V46BHHE  0.008556 0.006287 1.361000 0.828800 0.931700  0.379 0.534
V13RWB   0.008482 0.004846 1.750400 0.662400 1.170600  0.403 0.696
V49LYRE  0.007927 0.004849 1.634700 0.803100 0.000000  0.426 0.030 *
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V68PIL   0.007799 0.004781 1.631300 0.785200 0.000000  0.449 0.013 *
V80RF    0.007785 0.006023 1.292600 0.913500 0.304500  0.471 0.122
V54STHR  0.007684 0.004615 1.665200 0.758900 0.000000  0.493 0.004 **
V23RBFT  0.007670 0.005612 1.366800 0.883700 0.243500  0.516 0.704
V28VS    0.007670 0.007611 1.007700 0.709200 1.088900  0.538 0.640
V43GCU   0.007662 0.005636 1.359400 1.256100 0.583000  0.560 0.213
V16LR    0.007478 0.006026 1.241100 1.023200 1.479800  0.582 0.662
V20PCU   0.006977 0.006456 1.080800 1.268100 0.673000  0.602 0.372
V19ER    0.006769 0.006493 1.042500 0.266900 0.654600  0.621 0.753
V29CST   0.006698 0.006812 0.983200 0.314400 0.659900  0.641 0.487
V50CHE   0.006550 0.006045 1.083500 0.509600 0.384600  0.659 0.465
V35STP   0.006186 0.006202 0.997400 0.944200 1.011200  0.677 0.865
V42SIL   0.006120 0.004742 1.290500 1.363600 0.761200  0.695 0.921
V24BFCS  0.005843 0.005215 1.120300 0.496700 0.794400  0.712 0.413
V14AUR   0.005747 0.006056 0.948900 0.919800 0.987000  0.729 0.580
V55LHE   0.005677 0.005865 0.967900 0.590500 0.000000  0.745 0.262
V36YFHE  0.005166 0.005585 0.925100 1.134500 1.319700  0.760 0.976
V9SFW    0.005041 0.006178 0.815900 1.525000 1.081900  0.774 0.590
V64BELL  0.004931 0.008845 0.557500 0.000000 0.492000  0.789 0.362
V86BCOO  0.004867 0.005040 0.965600 0.481700 0.000000  0.803 0.083 .
V81SHBC  0.004290 0.004697 0.913300 1.191300 0.807100  0.815 0.953
V62RBTR  0.004175 0.005389 0.774800 0.281200 0.220000  0.827 0.182
V21ESP   0.003642 0.002780 1.310000 1.502100 1.346100  0.838 0.985
V96DF    0.003541 0.006353 0.557400 0.000000 0.349000  0.848 0.240
V4BTH    0.003496 0.002575 1.358000 2.294000 1.975000  0.858 0.948
V61SPW   0.003310 0.005938 0.557400 0.000000 0.326200  0.868 0.285
V8WNHE   0.003301 0.002559 1.290000 1.709100 1.582200  0.877 0.897
V85ROSE  0.003237 0.005792 0.558900 0.331700 0.000000  0.886 0.338
V7WEHE   0.002775 0.001747 1.588400 1.224500 1.318200  0.894 0.944
V27NHHE  0.002698 0.004842 0.557300 0.000000 0.256000  0.902 0.977
V58OBO   0.002685 0.004816 0.557400 0.000000 0.261700  0.910 0.870
V11CR    0.002557 0.001689 1.513300 1.710400 1.484400  0.917 0.991
V63DWS   0.002440 0.004377 0.557500 0.000000 0.243500  0.924 0.810
V57PINK  0.002440 0.004365 0.558900 0.250000 0.000000  0.931 0.464
V76GBB   0.002313 0.004139 0.558900 0.220000 0.000000  0.938 0.577
V66CBW   0.002257 0.004050 0.557400 0.000000 0.220000  0.945 0.599
V102KING 0.002240 0.004008 0.558900 0.236400 0.000000  0.951 0.334
V41SPP   0.002181 0.001351 1.614700 1.352200 1.526000  0.958 0.946
V51OWH   0.002147 0.003842 0.558900 0.220000 0.000000  0.964 0.607
V1GST    0.002076 0.002289 0.906700 1.379200 1.577800  0.970 0.524
V2EYR    0.002039 0.001545 1.319900 1.440100 1.316800  0.976 0.960
V15STTH  0.001903 0.001654 1.150800 1.857500 1.788200  0.981 0.990
V5GWH    0.001891 0.001382 1.367800 1.486100 1.309900  0.987 0.972
V3GF     0.001741 0.001033 1.685600 1.724600 1.569600  0.992 0.991
V6WTTR   0.001138 0.000630 1.805600 1.567100 1.579900  0.995 0.995
V56FTC   0.001005 0.000717 1.402700 1.231300 1.244700  0.998 0.998
V12LK    0.000762 0.000683 1.116400 1.324100 1.350100  1.000 0.996
V17WPHE  0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V25WAG   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V30BTR   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V31AMAG  0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V32SCC   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V34WSW   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V38GAL   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V39FHE   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V44MUSK  0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V45MGLK  0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V47RFC   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V52TRM   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V59YR    0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V60LFB   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V65LWB   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V70RSL   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V71PDOV  0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V72CRP   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
```

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V73JW    0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V74BCHE  0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V75RCR   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V77RRP   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V78LLOR  0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V79YTHE  0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V82AZKF  0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V84YRTH  0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V87LFC   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V88WG    0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V89PCOO  0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V90WTG   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V91NMIN  0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V92NFB   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V93DB    0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V94RBEE  0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V95HBC   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V97PCL   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V98FLAME 0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V99WWT   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V100WBWS 0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V101LCOR 0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
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Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Contrast: Montane Fore_Box-Ironbark

           average        sd     ratio       ava       avb cumsum     p
V18YTH    0.019614  0.002267  8.650000  0.000000  1.909900  0.042 0.002 **
V10WBSW   0.016737  0.002289  7.312000  1.622600  0.000000  0.079 0.005 **
V39FHE    0.015508  0.003007  5.157000  0.000000  1.532700  0.112 0.002 **
V21ESP    0.015277  0.002850  5.360000  1.502100  0.000000  0.145 0.005 **
V40BRTH   0.013739  0.008267  1.662000  0.000000  1.379300  0.175 0.012 *
V20PCU    0.013073  0.001670  7.829000  1.268100  0.000000  0.203 0.004 **
V26WWCH   0.012932  0.006735  1.920000  0.462400  1.655500  0.231 0.045 *
V13RWB    0.012500  0.008069  1.549000  0.662400  1.792200  0.258 0.070 .
V83SFC    0.012497  0.001221 10.232000  1.220000  0.000000  0.285 0.004 **
V44MUSK   0.011793  0.007641  1.543000  0.000000  1.180800  0.310 0.009 **
V58OBO    0.010960  0.002648  4.139000  0.000000  1.059200  0.334 0.002 **
V22SCR    0.010674  0.005558  1.920000  0.317500  1.350800  0.357 0.026 *
V46BHHE   0.010527  0.009610  1.095000  0.828800  1.798900  0.380 0.136
V75RCR    0.010410  0.003002  3.468000  0.000000  1.000900  0.402 0.001 ***
V37WHIP   0.010161  0.006135  1.656000  1.020400  0.000000  0.424 0.020 *
V56FTC    0.010047  0.005497  1.828000  1.231300  0.281200  0.446 0.041 *
V48YTBC   0.009965  0.006005  1.659000  0.958400  0.000000  0.468 0.024 *
V81SHBC   0.009699  0.005338  1.817000  1.191300  0.281200  0.489 0.021 *
V67GGC    0.009203  0.005719  1.609000  0.926600  0.000000  0.508 0.002 **
V80RF     0.009127  0.005635  1.620000  0.913500  0.000000  0.528 0.057 .
V16LR     0.008206  0.006541  1.255000  1.023200  0.576500  0.546 0.494
V11CR     0.008012  0.007131  1.124000  1.710400  0.992900  0.563 0.762
V49LYRE   0.007963  0.004911  1.621000  0.803100  0.000000  0.581 0.020 *
V4BTH     0.007900  0.002729  2.895000  2.294000  1.520900  0.598 0.690
V33RWH    0.007855  0.007080  1.109000  0.753700  1.512200  0.615 0.365
V23RBFT   0.007852  0.005901  1.331000  0.883700  0.320500  0.631 0.686
V68PIL    0.007835  0.004844  1.617000  0.785200  0.000000  0.648 0.012 *
V35STP    0.007792  0.006378  1.222000  0.944200  0.656700  0.665 0.592
V54STHR   0.007721  0.004680  1.650000  0.758900  0.000000  0.682 0.004 **
V28VS     0.007362  0.006716  1.096000  0.709200  0.971800  0.698 0.760
V19ER     0.007289  0.005572  1.308000  0.266900  0.846400  0.714 0.616
V36YFHE   0.006921  0.005444  1.271000  1.134500  1.383500  0.729 0.887
V66CBW    0.006194  0.006437  0.962000  0.000000  0.566600  0.742 0.026 *
V24BFCS   0.005995  0.005034  1.191000  0.496700  1.014400  0.755 0.378
V2EYR     0.005841  0.005810  1.005000  1.440100  0.874900  0.768 0.608
V90WTG    0.005838  0.006497  0.899000  0.000000  0.542000  0.780 0.107
V70RSL    0.005780  0.006013  0.961000  0.000000  0.557400  0.793 0.357
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V55LHE    0.005702  0.005915  0.964000  0.590500  0.000000  0.805 0.261
V61SPW    0.005691  0.005950  0.956000  0.000000  0.593000  0.817 0.054 .
V50CHE    0.005164  0.005411  0.954000  0.509600  0.000000  0.829 0.659
V86BCOO   0.004890  0.005087  0.961000  0.481700  0.000000  0.839 0.104
V43GCU    0.004840  0.005651  0.857000  1.256100  0.818400  0.850 0.860
V29CST    0.004252  0.005499  0.773000  0.314400  0.256000  0.859 0.964
V14AUR    0.004172  0.005415  0.771000  0.919800  1.211600  0.868 0.851
V3GF      0.004016  0.002231  1.800000  1.724600  1.330200  0.876 0.669
V79YTHE   0.003656  0.006555  0.558000  0.000000  0.396100  0.884 0.191
V30BTR    0.003534  0.006336  0.558000  0.000000  0.382900  0.892 0.808
V76GBB    0.003414  0.004520  0.755000  0.220000  0.210200  0.899 0.272
V5GWH     0.003383  0.002341  1.445000  1.486100  1.171900  0.907 0.803
V85ROSE   0.003252  0.005834  0.557000  0.331700  0.000000  0.914 0.345
V62RBTR   0.003172  0.005698  0.557000  0.281200  0.000000  0.920 0.368
V8WNHE    0.002763  0.002154  1.282000  1.709100  1.595400  0.926 0.941
V15STTH   0.002604  0.001970  1.321000  1.857500  1.627000  0.932 0.960
V57PINK   0.002450  0.004397  0.557000  0.250000  0.000000  0.937 0.493
V9SFW     0.002418  0.002010  1.203000  1.525000  1.472800  0.943 0.955
V94RBEE   0.002363  0.004237  0.558000  0.000000  0.256000  0.948 0.548
V96DF     0.002363  0.004237  0.558000  0.000000  0.256000  0.953 0.517
V102KING  0.002249  0.004035  0.557000  0.236400  0.000000  0.958 0.323
V7WEHE    0.002224  0.001655  1.344000  1.224500  1.296700  0.963 0.982
V51OWH    0.002157  0.003870  0.557000  0.220000  0.000000  0.967 0.607
V88WG     0.002119  0.003801  0.557000  0.000000  0.210200  0.972 0.523
V95HBC    0.002119  0.003801  0.557000  0.000000  0.210200  0.976 0.325
V12LK     0.002086  0.000765  2.726000  1.324100  1.122800  0.981 0.627
V63DWS    0.002031  0.003641  0.558000  0.000000  0.220000  0.985 0.928
V42SIL    0.002007  0.001772  1.133000  1.363600  1.255000  0.990 1.000
V1GST     0.001927  0.000613  3.143000  1.379200  1.425700  0.994 0.610
V41SPP    0.001767  0.001227  1.439000  1.352200  1.440100  0.998 0.945
V6WTTR    0.001142  0.000614  1.860000  1.567100  1.525100  1.000 0.992
V17WPHE   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V25WAG    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V27NHHE   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V31AMAG   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V32SCC    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V34WSW    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V38GAL    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V45MGLK   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V47RFC    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V52TRM    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V53MB     0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V59YR     0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V60LFB    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V64BELL   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V65LWB    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V69SKF    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V71PDOV   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V72CRP    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V73JW     0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V74BCHE   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V77RRP    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V78LLOR   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V82AZKF   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V84YRTH   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V87LFC    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V89PCOO   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V91NMIN   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V92NFB    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V93DB     0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V97PCL    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V98FLAME  0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V99WWT    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V100WBWS  0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V101LCOR  0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
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Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Contrast: Montane Fore_River Red Gu

           average        sd     ratio       ava       avb cumsum     p    
V4BTH     0.023240  0.001423 16.334000  2.294000  0.000000  0.032 0.001 ***
V32SCC    0.022619  0.002319  9.752000  0.000000  2.225000  0.062 0.002 ** 
V17WPHE   0.020648  0.001817 11.364000  0.000000  2.034900  0.090 0.007 ** 
V11CR     0.017360  0.001975  8.790000  1.710400  0.000000  0.114 0.004 ** 
V10WBSW   0.016529  0.001948  8.485000  1.622600  0.000000  0.136 0.005 ** 
V38GAL    0.016118  0.001999  8.065000  0.000000  1.582800  0.158 0.001 ***
V30BTR    0.015451  0.001552  9.955000  0.000000  1.520300  0.179 0.002 ** 
V21ESP    0.015090  0.002641  5.713000  1.502100  0.000000  0.200 0.003 ** 
V2EYR     0.014573  0.001010 14.428000  1.440100  0.000000  0.220 0.001 ***
V8WNHE    0.014442  0.006232  2.317000  1.709100  0.289600  0.239 0.003 ** 
V31AMAG   0.014171  0.001380 10.268000  0.000000  1.393900  0.258 0.002 ** 
V25WAG    0.013859  0.001151 12.039000  0.000000  1.365200  0.277 0.003 ** 
V41SPP    0.013725  0.001684  8.149000  1.352200  0.000000  0.296 0.002 ** 
V60LFB    0.013381  0.003234  4.138000  0.000000  1.312300  0.314 0.001 ***
V47RFC    0.012687  0.001289  9.845000  0.000000  1.251400  0.331 0.001 ***
V43GCU    0.012658  0.001321  9.581000  1.256100  0.000000  0.348 0.008 ** 
V7WEHE    0.012401  0.000859 14.431000  1.224500  0.000000  0.365 0.010 ** 
V26WWCH   0.012397  0.005637  2.199000  0.462400  1.578000  0.382 0.055 .  
V83SFC    0.012343  0.000888 13.906000  1.220000  0.000000  0.399 0.002 ** 
V45MGLK   0.012337  0.001123 10.986000  0.000000  1.216900  0.416 0.001 ***
V81SHBC   0.012054  0.001019 11.835000  1.191300  0.000000  0.432 0.003 ** 
V69SKF    0.011732  0.001094 10.720000  0.000000  1.156400  0.448 0.001 ***
V42SIL    0.011523  0.004695  2.455000  1.363600  0.442300  0.464 0.063 .  
V19ER     0.011415  0.005033  2.268000  0.266900  1.397300  0.479 0.025 *  
V36YFHE   0.011410  0.006858  1.664000  1.134500  0.000000  0.495 0.127    
V15STTH   0.011292  0.008100  1.394000  1.857500  0.745400  0.510 0.125    
V70RSL    0.010957  0.007153  1.532000  0.000000  1.085200  0.525 0.016 *  
V56FTC    0.010350  0.004225  2.450000  1.231300  0.220000  0.539 0.025 *  
V37WHIP   0.010040  0.006016  1.669000  1.020400  0.000000  0.552 0.025 *  
V48YTBC   0.009839  0.005880  1.673000  0.958400  0.000000  0.566 0.020 *  
V34WSW    0.009813  0.006345  1.546000  0.000000  0.961700  0.579 0.193    
V98FLAME  0.009699  0.006152  1.577000  0.000000  0.964700  0.592 0.010 ** 
V73JW     0.009395  0.005864  1.602000  0.000000  0.934100  0.605 0.001 ***
V67GGC    0.009094  0.005611  1.621000  0.926600  0.000000  0.618 0.005 ** 
V80RF     0.009018  0.005525  1.632000  0.913500  0.000000  0.630 0.066 .  
V52TRM    0.008836  0.005669  1.559000  0.000000  0.871300  0.642 0.034 *  
V6WTTR    0.008775  0.007475  1.174000  1.567100  0.736100  0.654 0.307    
V18YTH    0.008617  0.009023  0.955000  0.000000  0.845000  0.665 0.502    
V24BFCS   0.008611  0.005836  1.475000  0.496700  1.328700  0.677 0.027 *  
V93DB     0.008405  0.005073  1.657000  0.000000  0.829800  0.689 0.003 ** 
V91NMIN   0.007934  0.008238  0.963000  0.000000  0.767400  0.699 0.074 .  
V46BHHE   0.007914  0.008183  0.967000  0.828800  0.000000  0.710 0.667    
V49LYRE   0.007869  0.004819  1.633000  0.803100  0.000000  0.721 0.032 *  
V33RWH    0.007804  0.006307  1.237000  0.753700  0.532700  0.731 0.424    
V68PIL    0.007742  0.004750  1.630000  0.785200  0.000000  0.742 0.024 *  
V54STHR   0.007627  0.004585  1.663000  0.758900  0.000000  0.752 0.008 ** 
V20PCU    0.007523  0.006020  1.250000  1.268100  0.602800  0.762 0.192    
V94RBEE   0.007429  0.004516  1.645000  0.000000  0.732200  0.772 0.005 ** 
V5GWH     0.007231  0.004781  1.512000  1.486100  0.765000  0.782 0.256    
V28VS     0.007207  0.007247  0.994000  0.709200  0.686000  0.792 0.766    
V29CST    0.007165  0.004999  1.433000  0.314400  0.776000  0.802 0.307    
V35STP    0.007050  0.006207  1.136000  0.944200  1.638300  0.811 0.761    
V59YR     0.006644  0.006900  0.963000  0.000000  0.651800  0.820 0.020 *  
V23RBFT   0.006615  0.005151  1.284000  0.883700  0.456500  0.829 0.895    
V3GF      0.006534  0.001821  3.589000  1.724600  1.077300  0.838 0.437    
V13RWB    0.006293  0.007149  0.880000  0.662400  0.000000  0.847 0.951    
V16LR     0.006278  0.005139  1.222000  1.023200  1.252000  0.855 0.830    
V88WG     0.006250  0.006536  0.956000  0.000000  0.612900  0.864 0.022 *  
V90WTG    0.006250  0.006536  0.956000  0.000000  0.612900  0.872 0.073 .  
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V99WWT    0.006243  0.006502  0.960000  0.000000  0.615300  0.881 0.024 *
V58OBO    0.006193  0.006425  0.964000  0.000000  0.611700  0.889 0.234
V55LHE    0.005637  0.005827  0.967000  0.590500  0.000000  0.897 0.253
V72CRP    0.005399  0.005608  0.963000  0.000000  0.545600  0.904 0.018 *
V92NFB    0.005341  0.005564  0.960000  0.000000  0.513100  0.911 0.074 .
V71PDOV   0.005320  0.005756  0.924000  0.000000  0.530500  0.919 0.043 *
V50CHE    0.005101  0.005325  0.958000  0.509600  0.000000  0.926 0.638
V86BCOO   0.004831  0.005006  0.965000  0.481700  0.000000  0.932 0.106
V9SFW     0.004284  0.002149  1.993000  1.525000  1.145500  0.938 0.687
V12LK     0.004232  0.005626  0.752000  1.324100  0.910300  0.944 0.373
V14AUR    0.004098  0.005269  0.778000  0.919800  1.237700  0.949 0.867
V84YRTH   0.004003  0.007182  0.557000  0.000000  0.389600  0.955 0.209
V101LCOR  0.003592  0.006442  0.558000  0.000000  0.370000  0.960 0.193
V22SCR    0.003532  0.006323  0.559000  0.317500  0.000000  0.964 0.885
V85ROSE   0.003214  0.005752  0.559000  0.331700  0.000000  0.969 0.382
V62RBTR   0.003128  0.005600  0.559000  0.281200  0.000000  0.973 0.382
V87LFC    0.003051  0.005472  0.558000  0.000000  0.314400  0.977 0.606
V74BCHE   0.002765  0.004961  0.557000  0.000000  0.261700  0.981 0.180
V57PINK   0.002422  0.004335  0.559000  0.250000  0.000000  0.984 0.508
V76GBB    0.002296  0.004109  0.559000  0.220000  0.000000  0.987 0.594
V102KING  0.002224  0.003981  0.559000  0.236400  0.000000  0.990 0.315
V53MB     0.002160  0.003875  0.557000  0.000000  0.210200  0.993 0.874
V51OWH    0.002132  0.003815  0.559000  0.220000  0.000000  0.996 0.645
V95HBC    0.002101  0.003770  0.557000  0.000000  0.198800  0.999 0.348
V1GST     0.000715  0.000313  2.283000  1.379200  1.343300  1.000 0.999
V27NHHE   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V39FHE    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V40BRTH   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V44MUSK   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V61SPW    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V63DWS    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V64BELL   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V65LWB    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V66CBW    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V75RCR    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V77RRP    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V78LLOR   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V79YTHE   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V82AZKF   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V89PCOO   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V96DF     0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V97PCL    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V100WBWS  0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
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Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Contrast: Foothills Wo_Box-Ironbark

          average       sd    ratio      ava      avb cumsum     p
V39FHE   0.015833 0.002871 5.514000 0.000000 1.532700  0.047 0.002 **
V21ESP   0.014135 0.003111 4.543000 1.346100 0.000000  0.089 0.006 **
V44MUSK  0.012037 0.007747 1.554000 0.000000 1.180800  0.125 0.013 *
V75RCR   0.010634 0.002985 3.562000 0.000000 1.000900  0.157 0.001 ***
V56FTC   0.010423 0.005633 1.850000 1.244700 0.281200  0.188 0.033 *
V16LR    0.009727 0.006963 1.397000 1.479800 0.576500  0.217 0.182
V40BRTH  0.009298 0.009205 1.010000 1.423800 1.379300  0.245 0.195
V69SKF   0.009269 0.005598 1.656000 0.890600 0.000000  0.273 0.016 *
V46BHHE  0.009105 0.006329 1.439000 0.931700 1.798900  0.300 0.362
V53MB    0.008985 0.005474 1.641000 0.851600 0.000000  0.327 0.021 *
V58OBO   0.008774 0.004921 1.783000 0.261700 1.059200  0.353 0.029 *
V35STP   0.008296 0.006636 1.250000 1.011200 0.656700  0.378 0.427
V10WBSW  0.008277 0.008818 0.939000 0.773600 0.000000  0.402 0.615
V18YTH   0.007632 0.008144 0.937000 1.199300 1.909900  0.425 0.590
V81SHBC  0.007489 0.005740 1.305000 0.807100 0.281200  0.447 0.189
V20PCU   0.006999 0.007268 0.963000 0.673000 0.000000  0.468 0.344
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V19ER    0.006999 0.005848 1.197000 0.654600 0.846400  0.489 0.703
V29CST   0.006862 0.006603 1.039000 0.659900 0.256000  0.510 0.432
V43GCU   0.006594 0.005495 1.200000 0.583000 0.818400  0.529 0.496
V13RWB   0.006454 0.003518 1.834000 1.170600 1.792200  0.549 0.938
V11CR    0.006420 0.006540 0.982000 1.484400 0.992900  0.568 0.922
V28VS    0.006385 0.006620 0.964000 1.088900 0.971800  0.587 0.885
V61SPW   0.006374 0.006302 1.011000 0.326200 0.593000  0.606 0.020 *
V66CBW   0.006186 0.005875 1.053000 0.220000 0.566600  0.624 0.022 *
V26WWCH  0.006051 0.007359 0.822000 1.210300 1.655500  0.642 0.869
V90WTG   0.005966 0.006614 0.902000 0.000000 0.542000  0.660 0.105
V70RSL   0.005903 0.006116 0.965000 0.000000 0.557400  0.678 0.303
V9SFW    0.005785 0.005795 0.998000 1.081900 1.472800  0.695 0.465
V42SIL   0.005356 0.004742 1.130000 0.761200 1.255000  0.711 0.956
V2EYR    0.005255 0.005672 0.927000 1.316800 0.874900  0.727 0.745
V64BELL  0.005057 0.009076 0.557000 0.492000 0.000000  0.742 0.319
V4BTH    0.005028 0.003157 1.593000 1.975000 1.520900  0.757 0.852
V96DF    0.004848 0.006302 0.769000 0.349000 0.256000  0.771 0.071 .
V23RBFT  0.004574 0.005837 0.784000 0.243500 0.320500  0.785 0.983
V36YFHE  0.004518 0.002675 1.689000 1.319700 1.383500  0.798 0.981
V14AUR   0.004445 0.005204 0.854000 0.987000 1.211600  0.812 0.800
V50CHE   0.004163 0.007474 0.557000 0.384600 0.000000  0.824 0.794
V79YTHE  0.003726 0.006667 0.559000 0.000000 0.396100  0.835 0.173
V30BTR   0.003602 0.006445 0.559000 0.000000 0.382900  0.846 0.776
V63DWS   0.003558 0.004682 0.760000 0.243500 0.220000  0.857 0.674
V24BFCS  0.003325 0.004364 0.762000 0.794400 1.014400  0.867 0.893
V80RF    0.003296 0.005917 0.557000 0.304500 0.000000  0.876 0.902
V7WEHE   0.003045 0.002323 1.311000 1.318200 1.296700  0.885 0.911
V1GST    0.002847 0.001972 1.444000 1.577800 1.425700  0.894 0.145
V27NHHE  0.002771 0.004976 0.557000 0.256000 0.000000  0.902 0.967
V83SFC   0.002755 0.004946 0.557000 0.261700 0.000000  0.910 0.905
V15STTH  0.002694 0.001759 1.532000 1.788200 1.627000  0.918 0.959
V3GF     0.002586 0.002274 1.137000 1.569600 1.330200  0.926 0.944
V94RBEE  0.002409 0.004310 0.559000 0.000000 0.256000  0.933 0.538
V12LK    0.002400 0.001272 1.887000 1.350100 1.122800  0.941 0.547
V62RBTR  0.002317 0.004159 0.557000 0.220000 0.000000  0.947 0.593
V8WNHE   0.002311 0.000652 3.548000 1.582200 1.595400  0.954 0.981
V76GBB   0.002287 0.004092 0.559000 0.000000 0.210200  0.961 0.613
V22SCR   0.002258 0.001532 1.474000 1.364500 1.350800  0.968 0.971
V5GWH    0.002206 0.001744 1.265000 1.309900 1.171900  0.975 0.937
V88WG    0.002163 0.003870 0.559000 0.000000 0.210200  0.981 0.484
V95HBC   0.002163 0.003870 0.559000 0.000000 0.210200  0.987 0.257
V33RWH   0.001840 0.001210 1.520000 1.597500 1.512200  0.993 0.996
V41SPP   0.001677 0.001086 1.544000 1.526000 1.440100  0.998 0.957
V6WTTR   0.000709 0.000533 1.332000 1.579900 1.525100  1.000 0.999
V17WPHE  0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V25WAG   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V31AMAG  0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V32SCC   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V34WSW   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V37WHIP  0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V38GAL   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V45MGLK  0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V47RFC   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V48YTBC  0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V49LYRE  0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V51OWH   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V52TRM   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V54STHR  0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V55LHE   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V57PINK  0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V59YR    0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V60LFB   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V65LWB   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V67GGC   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V68PIL   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
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V71PDOV  0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V72CRP   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V73JW    0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V74BCHE  0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V77RRP   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V78LLOR  0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V82AZKF  0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V84YRTH  0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V85ROSE  0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V86BCOO  0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V87LFC   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V89PCOO  0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V91NMIN  0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V92NFB   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V93DB    0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V97PCL   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V98FLAME 0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V99WWT   0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V100WBWS 0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V101LCOR 0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
V102KING 0.000000 0.000000      NaN 0.000000 0.000000  1.000    NA
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Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Contrast: Foothills Wo_River Red Gu

           average        sd     ratio       ava       avb cumsum     p    
V32SCC    0.023093  0.001828 12.633000  0.000000  2.225000  0.035 0.001 ***
V17WPHE   0.021080  0.001241 16.985000  0.000000  2.034900  0.067 0.005 ** 
V4BTH     0.020489  0.002659  7.707000  1.975000  0.000000  0.098 0.002 ** 
V38GAL    0.016457  0.001736  9.477000  0.000000  1.582800  0.123 0.001 ***
V41SPP    0.015828  0.001134 13.956000  1.526000  0.000000  0.146 0.001 ***
V30BTR    0.015775  0.001205 13.087000  0.000000  1.520300  0.170 0.001 ***
V11CR     0.015410  0.001314 11.727000  1.484400  0.000000  0.194 0.005 ** 
V40BRTH   0.014624  0.009247  1.582000  1.423800  0.000000  0.216 0.007 ** 
V31AMAG   0.014468  0.001047 13.821000  0.000000  1.393900  0.238 0.001 ***
V25WAG    0.014149  0.000727 19.475000  0.000000  1.365200  0.259 0.004 ** 
V22SCR    0.014138  0.001253 11.280000  1.364500  0.000000  0.280 0.003 ** 
V21ESP    0.013956  0.002919  4.781000  1.346100  0.000000  0.301 0.014 *  
V7WEHE    0.013692  0.003128  4.377000  1.318200  0.000000  0.322 0.002 ** 
V2EYR     0.013677  0.002086  6.556000  1.316800  0.000000  0.343 0.002 ** 
V60LFB    0.013662  0.003176  4.302000  0.000000  1.312300  0.363 0.001 ***
V36YFHE   0.013654  0.002185  6.250000  1.319700  0.000000  0.384 0.024 *  
V8WNHE    0.013515  0.005982  2.259000  1.582200  0.289600  0.404 0.009 ** 
V47RFC    0.012953  0.001004 12.896000  0.000000  1.251400  0.424 0.001 ***
V45MGLK   0.012595  0.000795 15.834000  0.000000  1.216900  0.443 0.001 ***
V13RWB    0.012138  0.001836  6.609000  1.170600  0.000000  0.461 0.085 .  
V70RSL    0.011186  0.007252  1.543000  0.000000  1.085200  0.478 0.006 ** 
V15STTH   0.011042  0.008000  1.380000  1.788200  0.745400  0.495 0.180    
V33RWH    0.011004  0.005906  1.863000  1.597500  0.532700  0.511 0.031 *  
V56FTC    0.010724  0.004330  2.477000  1.244700  0.220000  0.528 0.017 *  
V34WSW    0.010020  0.006434  1.557000  0.000000  0.961700  0.543 0.205    
V98FLAME  0.009901  0.006236  1.588000  0.000000  0.964700  0.558 0.008 ** 
V46BHHE   0.009696  0.006173  1.571000  0.931700  0.000000  0.573 0.242    
V73JW     0.009590  0.005943  1.614000  0.000000  0.934100  0.587 0.001 ***
V18YTH    0.009296  0.008040  1.156000  1.199300  0.845000  0.601 0.371    
V52TRM    0.009021  0.005747  1.570000  0.000000  0.871300  0.615 0.028 *  
V6WTTR    0.008899  0.007759  1.147000  1.579900  0.736100  0.628 0.297    
V93DB     0.008581  0.005139  1.670000  0.000000  0.829800  0.641 0.001 ***
V42SIL    0.008525  0.004802  1.776000  0.761200  0.442300  0.654 0.468    
V81SHBC   0.008439  0.005218  1.617000  0.807100  0.000000  0.667 0.081 .  
V10WBSW   0.008169  0.008666  0.943000  0.773600  0.000000  0.679 0.660    
V91NMIN   0.008103  0.008379  0.967000  0.000000  0.767400  0.691 0.076 .  
V19ER     0.007916  0.006665  1.188000  0.654600  1.397300  0.703 0.452    
V53MB     0.007757  0.005181  1.497000  0.851600  0.210200  0.715 0.056 .  
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V28VS     0.007735  0.007384  1.048000  1.088900  0.686000  0.727 0.634
V94RBEE   0.007585  0.004574  1.658000  0.000000  0.732200  0.738 0.003 **
V20PCU    0.007072  0.006526  1.084000  0.673000  0.602800  0.749 0.323
V29CST    0.006866  0.005049  1.360000  0.659900  0.776000  0.759 0.472
V59YR     0.006784  0.007017  0.967000  0.000000  0.651800  0.769 0.006 **
V35STP    0.006652  0.006625  1.004000  1.011200  1.638300  0.779 0.801
V88WG     0.006382  0.006647  0.960000  0.000000  0.612900  0.789 0.007 **
V90WTG    0.006382  0.006647  0.960000  0.000000  0.612900  0.799 0.052 .
V99WWT    0.006373  0.006612  0.964000  0.000000  0.615300  0.808 0.010 **
V58OBO    0.006330  0.006124  1.034000  0.261700  0.611700  0.818 0.235
V26WWCH   0.005950  0.006536  0.910000  1.210300  1.578000  0.827 0.887
V43GCU    0.005949  0.006303  0.944000  0.583000  0.000000  0.836 0.703
V9SFW     0.005874  0.004177  1.406000  1.081900  1.145500  0.845 0.495
V5GWH     0.005644  0.004760  1.186000  1.309900  0.765000  0.853 0.550
V72CRP    0.005509  0.005702  0.966000  0.000000  0.545600  0.862 0.014 *
V24BFCS   0.005494  0.005150  1.067000  0.794400  1.328700  0.870 0.493
V92NFB    0.005455  0.005659  0.964000  0.000000  0.513100  0.878 0.090 .
V71PDOV   0.005431  0.005852  0.928000  0.000000  0.530500  0.886 0.024 *
V3GF      0.005121  0.001989  2.575000  1.569600  1.077300  0.894 0.523
V64BELL   0.004995  0.008940  0.559000  0.492000  0.000000  0.902 0.348
V23RBFT   0.004889  0.004924  0.993000  0.243500  0.456500  0.909 0.984
V12LK     0.004650  0.005759  0.808000  1.350100  0.910300  0.916 0.289
V14AUR    0.004157  0.005215  0.797000  0.987000  1.237700  0.922 0.846
V50CHE    0.004108  0.007354  0.559000  0.384600  0.000000  0.928 0.793
V84YRTH   0.004088  0.007314  0.559000  0.000000  0.389600  0.935 0.166
V69SKF    0.003666  0.005226  0.701000  0.890600  1.156400  0.940 0.807
V101LCOR  0.003664  0.006556  0.559000  0.000000  0.370000  0.946 0.139
V96DF     0.003588  0.006422  0.559000  0.349000  0.000000  0.951 0.225
V61SPW    0.003353  0.006002  0.559000  0.326200  0.000000  0.956 0.219
V80RF     0.003252  0.005822  0.559000  0.304500  0.000000  0.961 0.908
V87LFC    0.003113  0.005569  0.559000  0.000000  0.314400  0.966 0.641
V74BCHE   0.002825  0.005054  0.559000  0.000000  0.261700  0.970 0.146
V27NHHE   0.002735  0.004896  0.559000  0.256000  0.000000  0.974 0.968
V83SFC    0.002720  0.004869  0.559000  0.261700  0.000000  0.978 0.914
V16LR     0.002626  0.002077  1.264000  1.479800  1.252000  0.982 0.987
V1GST     0.002484  0.002388  1.040000  1.577800  1.343300  0.986 0.309
V63DWS    0.002472  0.004425  0.559000  0.243500  0.000000  0.990 0.803
V62RBTR   0.002287  0.004094  0.559000  0.220000  0.000000  0.993 0.599
V66CBW    0.002287  0.004094  0.559000  0.220000  0.000000  0.997 0.642
V95HBC    0.002146  0.003840  0.559000  0.000000  0.198800  1.000 0.286
V37WHIP   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V39FHE    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V44MUSK   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V48YTBC   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V49LYRE   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V51OWH    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V54STHR   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V55LHE    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V57PINK   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V65LWB    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V67GGC    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V68PIL    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V75RCR    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V76GBB    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V77RRP    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V78LLOR   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V79YTHE   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V82AZKF   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V85ROSE   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V86BCOO   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V89PCOO   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V97PCL    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V100WBWS  0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V102KING  0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Contrast: Box-Ironbark_River Red Gu

           average        sd     ratio       ava       avb cumsum     p
V32SCC    0.023209  0.002485  9.340000  0.000000  2.225000  0.036 0.003 **
V17WPHE   0.021185  0.001961 10.802000  0.000000  2.034900  0.068 0.008 **
V46BHHE   0.018779  0.002092  8.977000  1.798900  0.000000  0.097 0.001 ***
V13RWB    0.018603  0.002860  6.505000  1.792200  0.000000  0.126 0.001 ***
V38GAL    0.016539  0.002119  7.804000  0.000000  1.582800  0.151 0.001 ***
V4BTH     0.015839  0.002532  6.255000  1.520900  0.000000  0.175 0.014 *
V39FHE    0.015712  0.002885  5.447000  1.532700  0.000000  0.200 0.002 **
V41SPP    0.014960  0.001224 12.222000  1.440100  0.000000  0.222 0.001 ***
V31AMAG   0.014541  0.001484  9.795000  0.000000  1.393900  0.245 0.003 **
V36YFHE   0.014477  0.004922  2.941000  1.383500  0.000000  0.267 0.012 *
V25WAG    0.014220  0.001253 11.353000  0.000000  1.365200  0.289 0.004 **
V22SCR    0.014102  0.002784  5.066000  1.350800  0.000000  0.311 0.002 **
V40BRTH   0.013917  0.008320  1.673000  1.379300  0.000000  0.332 0.010 **
V8WNHE    0.013741  0.005715  2.405000  1.595400  0.289600  0.353 0.010 **
V60LFB    0.013731  0.003352  4.097000  0.000000  1.312300  0.374 0.001 ***
V7WEHE    0.013625  0.003185  4.278000  1.296700  0.000000  0.395 0.003 **
V47RFC    0.013017  0.001371  9.494000  0.000000  1.251400  0.415 0.002 **
V45MGLK   0.012658  0.001205 10.507000  0.000000  1.216900  0.435 0.002 **
V30BTR    0.012531  0.007039  1.780000  0.382900  1.520300  0.454 0.009 **
V69SKF    0.012038  0.001175 10.245000  0.000000  1.156400  0.472 0.002 **
V44MUSK   0.011946  0.007696  1.552000  1.180800  0.000000  0.491 0.006 **
V18YTH    0.011253  0.009055  1.243000  1.909900  0.845000  0.508 0.097 .
V42SIL    0.011190  0.003985  2.808000  1.255000  0.442300  0.525 0.080 .
V35STP    0.010880  0.007777  1.399000  0.656700  1.638300  0.542 0.072 .
V75RCR    0.010550  0.002971  3.552000  1.000900  0.000000  0.558 0.001 ***
V33RWH    0.010203  0.005935  1.719000  1.512200  0.532700  0.574 0.052 .
V34WSW    0.010070  0.006524  1.544000  0.000000  0.961700  0.589 0.168
V11CR     0.009963  0.006247  1.595000  0.992900  0.000000  0.605 0.384
V98FLAME  0.009949  0.006321  1.574000  0.000000  0.964700  0.620 0.009 **
V73JW     0.009637  0.006026  1.599000  0.000000  0.934100  0.635 0.002 **
V15STTH   0.009594  0.007905  1.214000  1.627000  0.745400  0.649 0.392
V52TRM    0.009066  0.005827  1.556000  0.000000  0.871300  0.663 0.018 *
V2EYR     0.009033  0.005592  1.615000  0.874900  0.000000  0.677 0.126
V93DB     0.008624  0.005214  1.654000  0.000000  0.829800  0.690 0.001 ***
V70RSL    0.008584  0.006538  1.313000  0.557400  1.085200  0.704 0.059 .
V6WTTR    0.008576  0.007595  1.129000  1.525100  0.736100  0.717 0.351
V43GCU    0.008275  0.004993  1.657000  0.818400  0.000000  0.730 0.122
V91NMIN   0.008145  0.008464  0.962000  0.000000  0.767400  0.742 0.064 .
V16LR     0.007429  0.006376  1.165000  0.576500  1.252000  0.753 0.645
V28VS     0.007148  0.006803  1.051000  0.971800  0.686000  0.764 0.780
V29CST    0.006978  0.005342  1.306000  0.256000  0.776000  0.775 0.392
V59YR     0.006818  0.007086  0.962000  0.000000  0.651800  0.786 0.011 *
V58OBO    0.006742  0.004954  1.361000  1.059200  0.611700  0.796 0.160
V90WTG    0.006683  0.006034  1.107000  0.542000  0.612900  0.806 0.039 *
V94RBEE   0.006671  0.005063  1.318000  0.256000  0.732200  0.817 0.011 *
V88WG     0.006408  0.005997  1.069000  0.210200  0.612900  0.826 0.011 *
V99WWT    0.006405  0.006678  0.959000  0.000000  0.615300  0.836 0.010 **
V66CBW    0.006280  0.006500  0.966000  0.566600  0.000000  0.846 0.029 *
V19ER     0.006192  0.005991  1.034000  0.846400  1.397300  0.855 0.827
V20PCU    0.006104  0.006440  0.948000  0.000000  0.602800  0.865 0.675
V61SPW    0.005762  0.006007  0.959000  0.593000  0.000000  0.874 0.038 *
V23RBFT   0.005699  0.005229  1.090000  0.320500  0.456500  0.882 0.954
V72CRP    0.005535  0.005754  0.962000  0.000000  0.545600  0.891 0.012 *
V92NFB    0.005483  0.005717  0.959000  0.000000  0.513100  0.899 0.083 .
V71PDOV   0.005457  0.005906  0.924000  0.000000  0.530500  0.908 0.026 *
V5GWH     0.004967  0.004736  1.049000  1.171900  0.765000  0.915 0.691
V9SFW     0.004265  0.002744  1.554000  1.472800  1.145500  0.922 0.699
V84YRTH   0.004108  0.007375  0.557000  0.000000  0.389600  0.928 0.162
V56FTC    0.003830  0.004875  0.786000  0.281200  0.220000  0.934 0.969
V12LK     0.003782  0.004861  0.778000  1.122800  0.910300  0.940 0.417
```

```
V79YTHE   0.003700  0.006623  0.559000  0.396100  0.000000  0.946 0.177
V101LCOR  0.003681  0.006605  0.557000  0.000000  0.370000  0.951 0.119
V24BFCS   0.003288  0.002264  1.452000  1.014400  1.328700  0.956 0.880
V95HBC    0.003249  0.004283  0.759000  0.210200  0.198800  0.961 0.039 *
V3GF      0.003169  0.001979  1.601000  1.330200  1.077300  0.966 0.857
V87LFC    0.003127  0.005611  0.557000  0.000000  0.314400  0.971 0.598
V74BCHE   0.002840  0.005099  0.557000  0.000000  0.261700  0.975 0.141
V81SHBC   0.002627  0.004702  0.559000  0.281200  0.000000  0.979 0.995
V96DF     0.002392  0.004281  0.559000  0.256000  0.000000  0.983 0.509
V76GBB    0.002269  0.004061  0.559000  0.210200  0.000000  0.986 0.595
V53MB     0.002217  0.003979  0.557000  0.000000  0.210200  0.990 0.880
V63DWS    0.002056  0.003679  0.559000  0.220000  0.000000  0.993 0.931
V1GST     0.001950  0.001137  1.714000  1.425700  1.343300  0.996 0.596
V26WWCH   0.001348  0.001039  1.297000  1.655500  1.578000  0.998 0.987
V14AUR    0.001268  0.000601  2.111000  1.211600  1.237700  1.000 0.993
V10WBSW   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V21ESP    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V27NHHE   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V37WHIP   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V48YTBC   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V49LYRE   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V50CHE    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V51OWH    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V54STHR   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V55LHE    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V57PINK   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V62RBTR   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V64BELL   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V65LWB    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V67GGC    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V68PIL    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V77RRP    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V78LLOR   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V80RF     0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V82AZKF   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V83SFC    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V85ROSE   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V86BCOO   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V89PCOO   0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V97PCL    0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V100WBWS  0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
V102KING  0.000000  0.000000       NaN  0.000000  0.000000  1.000    NA
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
Permutation: free
Number of permutations: 999
```

Extract one comparison

The output can be long. The following code extracts the first pairwise habitat comparison.

CODE

```
first_comparison <- summary(Birdsim)[[1]]
head(first_comparison, 10)
```

OUTPUT

```
           average          sd    ratio       ava       avb     cumsum     p
V17WPHE 0.01977874 0.011557166 1.711384 0.6019704 2.2477343 0.04046870 0.003
V34WSW  0.01897905 0.007112357 2.668462 0.2475148 1.8559931 0.07930117 0.001
```

```
V11CR   0.01685036 0.007496274 2.247831 1.4434071 0.0000000 0.11377819 0.001
V6WTTR  0.01545551 0.006359050 2.430475 1.3258152 0.0000000 0.14540126 0.002
V15STTH 0.01493983 0.009011889 1.657791 1.4754269 0.3233922 0.17596921 0.008
V25WAG  0.01396604 0.005801616 2.407268 0.2050889 1.3759763 0.20454472 0.002
V27NHHE 0.01276927 0.008435386 1.513774 0.6891770 1.6462619 0.23067153 0.044
V3GF    0.01248274 0.008487765 1.470674 1.5693228 0.6464682 0.25621210 0.001
V4BTH   0.01147159 0.009943672 1.153658 1.9464092 1.1842313 0.27968378 0.129
V36YFHE 0.01070821 0.008005858 1.337547 1.1213827 1.2437450 0.30159353 0.086
```

Interpretation

Species with larger average contributions are those that most strongly separate two habitat types.

These may be habitat-associated species, but SIMPER should be interpreted carefully because common and abundant species often contribute strongly.

Questions for students

Question 1

What does it mean if two sites are close together in the nMDS plot?

Answer guide

They have similar bird community composition based on Bray-Curtis dissimilarity.

Question 2

Why did we transform the bird abundance data before calculating Bray-Curtis dissimilarities?

Answer guide

The transformation reduces the dominance of very abundant species and allows less abundant species to contribute more to the multivariate pattern.

Question 3

What does a significant PERMANOVA tell us?

Answer guide

It tells us that bird community composition differs among habitat types more than expected by chance.

Question 4

Why should we check dispersion with `betadisper()`?

Answer guide

Because PERMANOVA can be influenced by differences in the spread of groups, not only differences in their centres.

Question 5

What does SIMPER add to the analysis?

Answer guide

SIMPER helps identify which species contribute most to the observed differences among habitat types.

Key take-home message

Hierarchical clustering, nMDS, ANOSIM, PERMANOVA and SIMPER all use the same basic idea: sites can be compared based on their whole community composition.

For the Mac Nally bird data, these tools let us ask whether forest and woodland habitat types support distinct bird assemblages, and which species contribute most to those differences.